
* TS-2068 LOW LEVEL UNIFIED COMMUNICATION PROTOCOL VERS: 2.4 AGO-2023 *

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**** PRELIMINAR ****
**** AND ****
**** CONFIDENTIAL ****

INTRODUCTION

This document details the communication protocol specifications between the operating system of the TS-2068 computer and the peripherals that support functionalities of storage and communication.

The new basic commands added to the traditional ones use the protocol TPI ("Timex Protocol Interface"). In this document you will find examples of these extended commands in Basic (called TPI commands from now on) and the detail of the low-level protocol data exchange for each one of them.

The first part of this document, corresponds to the TPI storage commands such as (SAVE, LOAD, VERIFY, MERGE), and to the TPI printer commands (COPY, LPRINT, etc).

In this document you will also find the communication messages used to load and save the entire memory of the TS2068 to a "SNAPSHOT" file in .TMS format called "Timex Memory Snapshot".

The document also includes the messages used to be able to run other operating systems written for this hardware platform. As an example, we can mention CP/M in its 2.2 or 3.0 versions.

The latest available version of this TPI command extension for the TS-2068, is included in the original TS-PICO interface launched this year for the 40th Anniversary of the TS-2068.

For more details on the use of the commands and functionalities, please consult the TS-PICO user manual which explains with examples, the implementation of this protocol in the distribution version.

The information herein is confidential, only the people authorized may use it in order to develop the interfaces and firmware required, and must not be distributed or published to third parties without the previous express author's consent.

The author is not responsible for any damage or harm caused by the use of the information contained at this document. However, if you find any error or consider that a functionality should be added or improved, please contact me at panegustavo@yahoo.com.ar with the subject TPI ERROR or TPI CHANGE REQUEST (these observations will be analyzed and could be included at further releases of this document).

This document supersedes all previous versions and may change without notice.

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INTRODUCTION:

THE COMMUNICATIONS PROTOCOL BETWEEN TS-2068 AND TS-PICO,
USES 2 Z80 I/O BIDIRECTIONAL COMMUNICATION PORTS:

(ADDRESS 14) DATA PORT (D0 - D7 8 BITS DATA)

(ADDRESS 15) CONTROL PORT (D6 = CONTINUE FLAG)

D6=0 MEANS BUSY (PROCESSING MESSAGE)

D6=1 MEANS CONTINUE (RETURN CONTROL TO THE ROM)

WHEN THE SYSTEM BOOTS UP, THE COMMUNICATION PORTS ARE INITIALIZED TO THE
VALUE OF 00h IN ALL THE BITS.

THE SYSTEM COMMUNICATES THROUGH A DATA BLOCK WITH MESSAGES PREDEFINED AND
CONTROLLES THEIR INTEGRITY BY A CRC VALUE SENT AS LAST BYTE. (SEE MESSAGES
DESCRIPTION CHAPTER)

ONCE THE CRC IS SENT, IT MUST BE COMPARED WITH THE LOCAL VALUE CALCULATED
FROM THE BEGINNING OF THE MESSAGE AND IF THEY MATCH, THE MESSAGE IS VALID,
AND THE RESPONSE SHOULD BE 01 (WHICH MEANS OK). OTHERWISE, YOU SHOULD
RESPOND WITH THE CORRESPONDING ERROR CODE (SEE TABLE OF ERRORS).

AS A GENERAL RULE, THE VALUE OF THE ERROR CODE CAN BE CALCULATED AS THE CRC
OF ALL BYTES RECEIVED +01

IF THE RESULTING FINAL VALUE IS 0x01, NO INFORMATION TRANSMISSION ERROR HAS
OCCURRED AND YOU SHOULD PROCEED WITH THE EXECUTION OF THE COMMAND NORMALLY.

IF THE VALUE IS DIFFERENT, THE VALUE RETURNED SHOULD BE A VALID ERROR CODE
ACCORDING TO THE ERROR TABLE OPTIONS.

IT IS IMPORTANT TO CARRY OUT THE CRC CONTROL TO BE ABLE TO DETECT INCOMPLETE
MESSAGES, WRONG MESSAGES, NOT SUPPORTED MESSAGES, MESSAGES WITH TRANSMISSION
FAILURES, OVERRUN ERRORS OR LACK OF SYNCHRONISM. IN ALL THESE CASES, THE
RECEIVER BUFFER MUST BE DISCARDED. THE TS-PICO VERSION DOES NOT SUPPORT
MESSAGE RETRANSMISSION, THEREFORE IT IS NOT RECOMMENDED TO RESPOND WITH A
STATIC VALUE (Status Code 0x01 = OK) YOU SHOULD ALWAYS RESPOND WITH THE
CORRECT CODE ACCORDING TO THE CASE, TO AVOID LOSS OF INFORMATION OR
MALFORMED FILES.

FOR THE SLVM COMMANDS 2 BYTES WITH THE SESSION ID ARE SENT. THIS SESSION ID
SHOULD BE THE SAME IN THE HEADER BLOCK AND THE DATA BLOCK. IF THE INFORMATION
OF SESSION CONTAINS VALUE 0000, THE MESSAGE BLOCK MEANS THAT IT WAS NOT GENERATED
FROM A BASIC COMMAND.

BELOW IS AN EXAMPLE OF THE TYPICAL BASIC SLVM GROUP (SAVE).

THE COMPLETE COMMAND INCLUDES THE FOLLOWING PHASES:

- 1) PRE-HEADER MESSAGE
- 2) RETURN ERROR CODE
- 3) WAIT FOR EXECUTION
- 4) HEADER BLOCK
- 5) WAIT FOR EXECUTION
- 6) RETURN ERROR CODE
- 7) DATA BLOCK
- 8) WAIT FOR EXECUTION
- 9) RETURN ERROR CODE

IN THE PHASE OF WAIT FOR EXECUTION A 2.8ms TIMER IS ACTIVATED. WHILE THE
TS-2068 IS WAITING THE CONTINUE SIGNAL, READING THE BIT 6 AT PORT 0x0F,
THE TS-PICO SHOULD EXECUTE THE COMMAND AND RESPOND DURING THAT PERIOD OF TIME.
IN CASE THE TIMER EXPIRES WAITING FOR THE CONTINUE SIGNAL, AN ERROR
J - INVALID I/O DEVICE (TIMEOUT CONDITION) IS RETURNED AUTOMATICALLY.

141
142 MENSAJE DE ENVIO DE CABECERA DE COMMANDOS
143 -----
144

145 =====

146	BYTE	STAT	DATA	DATA	BYTE DESCRIPTION
147	NUMB	IN	OUT	IN	
148	----	----	----	----	
149	PORT	RD-15	WR-14	RD-14	Z80 I/O PORT
150	=====				
151	00		0x00		Send BLOCK TYPE
152	01		0x00		Send TADDR (00 = SAVE)
153	02		0xFF		Send BYTE 1 *CMD DEPENDANT*
154	03		0x63		Send BYTE 2 *CMD DEPENDANT*
155	04		0x10		Send BYTE 3 *CMD DEPENDANT*
156	05		0xA4		Send BYTE 4 *CMD DEPENDANT*
157	06		0x6C		Send BYTE 5 *CMD DEPENDANT*
158	07		0x11		Send BYTE 6 *CMD DEPENDANT*
159	08		0x00		Send BYTE 7 *CMD DEPENDANT*
160	-----				
161	09		0x55		Send CRC FROM TIMEX (from bytes 00 to 08)
162	-----				
163	10			0x01	Read Status Code (01 = No Error)
164	-----				
165	-----WAIT EXECUTION 2.8ms-----				
166	-----				
167	11	0x40			Read CONTINUE FLAG
168	-----				

169
170 2ND BLOCK OF THE COMMAND
171 -----

172 =====

173	BYTE	STAT	DATA	DATA	DESCRIPTION
174	NUMB	IN	OUT	IN	
175	----	----	----	----	
176	PORT	RD-15	WR-14	RD-14	Z80 I/O PORT
177	=====				
178	-----				
179	12		0x44		"D" (DATA BLOCK CMD)
180	13		0x0A		COMND LEN L
181	14		0x00		COMND LEN H
182	15		0x54		"T"
183	16		0x50		"P"
184	17		0x49		"I"
185	18		0x3A		": "
186	19		0x44		"D"
187	20		0x45		"E"
188	21		0x4C		"L"
189	22		0x45		"E"
190	23		0x54		"T"
191	24		0x45		"E"
192	-----				
193	25		0x20		CRC FROM TIMEX
194	-----				
195	-----WAIT EXECUTION 2.8ms-----				
196	-----				
197	26	0x40			CONTINUE FLAG
198	-----				
199	27			0x01	Status Code (No Error)
200	-----				

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211
212 FIELD BLOCK TYPE
213 -----
214

215 VALUE DESCRIPTION
216 -----

217 0x00 PRE-HEADER BLOCK MESSAGE (SLVM OPERATION)
218 0xFF DATA-BLOCK MESSAGE (SLVM OPERATION)
219 0x42 "A" (ASM COMND) * RESERVED - NOT USED IN CURRENT TS-PICO*
220 0x42 "B" (BASIC COMMAND BLOCK)
221 0x43 "C" (CP/M CBIOS COMND)
222 0x44 "D" (DATA BLOCK)
223 0x45 "E" (EXTRA DATA) * RESERVED - NOT USED IN CURRENT TS-PICO*
224

225 FIELD COMMAND
226 TADDR ID
227 -----

228 0x00 SAVE
229 0x01 LOAD
230 0x02 VERIFY
231 0x03 MERGE
232 0x04 COPY
233 0x05 LPRINT
234 0x06 NEW * RESERVED - NOT USED IN CURRENT TS-PICO *
235 0x07 RUN * RESERVED - NOT USED IN CURRENT TS-PICO *
236 0x08 SET SNAP FILENAME
237 0x09 SAVE SNAP
238 0x0A LOAD SNAP
239

240 FIELD STATUS CODE
241 -----
242

243 ERROR TS-2068 BASIC
244 CODE EXECUTION
245 ;-----

246 0x00 PRINT REPORT J - INVALID I/O DEVICE
247 0x01 PRINT OK
248 0x02 PRINT REPORT R - TAPE LOADING ERROR
249 0x03 PRINT REPORT F - INVALID FILE NAME
250 0x04 PRINT REPORT Q - PARAMETER ERROR
251 0x05 PRINT REPORT C - NON SENSE IN BASIC
252 0x06 PRINT REPORT 6 - NUMBER TOO BIG
253 0x07 PRINT REPORT 8 - END OF FILE
254 0x08 PRINT REPORT A - INVALID ARGUMENT
255 0x09 PRINT REPORT 9 - STOP
256 0x0A PRINT REPORT J - INVALID I/O DEVICE
257 +0x0A -> 0x7F PRINT REPORT D - BREAK-CONT REPEATS
258 080H * FRAME RETRANSMISSION - NOT USED IN CURRENT TS-PICO BASIC OS*
259 081H FUNCTION PRINT STRING IN MAIN SCREEN
260 082H FUNCTION PRINT STRING IN MAIN SCREEN & RETURN A KEY
261 083H FUNCTION PRINT CHARACTER
262 084H FUNCTION RETURN KEY
263 085H FUNCTION GET STATUS (B0 = KEYBOARD / B1 = AUX / B2 = PRINTER / B3 = DISK)
264 086H FUNCTION PRINT STRING WITH LOOP (Y/N)
265 087H -> 0xFF * RESERVED - NOT USED IN CURRENT TS-PICO *
266
267
268

269 USING STATUS CODE > 080H
270 -----
271

272 AT THE LAST PHASE OF THE COMMAND, IF THE STATUS CODE RECEIVED IS GREATER
273 THAN 080H, IT MEANS THAT ADDITIONAL INFORMATION IS REQUIRED BY THE FUNCTION
274 AND SHOULD BE ADDED TO THE COMMUNICATION.
275

276 THE ADDITIONAL INFORMATION VARIES DEPENDING ON THE FUNCTION YOU WANT TO EXECUTE.
277

278 BELOW IS A DETAIL OF THE DATA FLOW CORRESPONDING TO THE ADDITIONAL FUNCTIONS:
279
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281
282 FUNCTION PRINT STRING IN MAIN SCREEN (081H)
283 -----

284
285 THE PRINT_STRING FUNCTION (0X81H) MUST BE CALLED ONLY AT THE LAST STATUS CODE
286 OF THE COMMAND.
287

288 THE NEXT VALUE TO SEND IS THE ERROR CODE THAT SHOULD BE SENT BACK TO THE OPERATING
289 SYSTEM, ACCORDING TO THE TABLE OF VALID RETURN CODES. (Status Code 01 = No Error).
290

291 THE STRING TO PRINT ON THE MAIN SCREEN IS SENT CHARACTER BY CHARACTER FINISHING
292 WITH THE NULL CHARACTER.
293

294 THEN THE ERROR CODE SHOULD BE SENT BACK TO THE OPERATING SYSTEM ACCORDING TO THE
295 TABLE OF RETURN CODES. GENERALLY, IF NO ERRORS OCCURED, THE VALUE SHOULD BE 0x01
296 (Status Code 01 = No Error)
297

298 =====
299 BYTE STAT DATA DATA DESCRIPTION
300 NUMB IN OUT IN
301 -----
302 PORT RD-15 WR-14 RD-14 Z80 I/O PORT
303 =====
304 .
305 .
306 .
307 .

308 -----
309 25 0x20 Send CRC FROM TIMEX
310 -----
311 -----WAIT EXECUTION 2.8ms-----
312 -----
313 26 0x40 Read CONTINUE FLAG
314 -----
315 27 0x81 Read Status Code (81 = PRINT_STRING)
316 -----
317 28 0x01 Return Status Code (01 = No Error)
318 -----
319 29 0x41 Character ASCII to Print
320 30 0x42 Character ASCII to Print
321 31 0x43 Character ASCII to Print
322 32 0x44 Character ASCII to Print
323 33 0x45 Character ASCII to Print
324 34 0x00 End of String 00 (NULL)
325 -----

326
327
328 FUNCTION PRINT STRING IN MAIN SCREEN & RETURN A KEY (082H)
329 -----

330 THE FUNCTION PRINT_STRING & RETURN A KEY (0X82H) MUST BE CALLED ONLY AT THE LAST
331 STATUS CODE OF THE COMMAND.
332

333 THE NEXT VALUE TO SEND IS THE ERROR CODE THAT SHOULD BE SENT BACK TO THE OPERATING
334 SYSTEM, ACCORDING TO THE TABLE OF VALID RETURN CODES. (Status Code 01 = No Error).
335

336 THE STRING TO PRINT ON THE MAIN SCREEN IS SENT CHARACTER BY CHARACTER FINISHING
337 WITH THE NULL CHARACTER. THEN, THE SYSTEM WAITS TILL A KEY IS PRESSED, AND THE
338 ASCII CODE OF THE KEY IS SENT TO THE TS-PICO.
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351
352 THEN THE ERROR CODE SHOULD BE SENT BACK TO THE OPERATING SYSTEM ACCORDING TO THE
353 TABLE OF RETURN CODES.
354

355 =====

356 BYTE	STAT	DATA	DATA	DESCRIPTION
357 NUMB	IN	OUT	IN	
358 ----	----	----	----	-----
359 PORT	RD-15	WR-14	RD-14	Z80 I/O PORT

360 =====

361 .

362 -----

363 25		0x20		Send CRC FROM TIMEX
--------	--	------	--	---------------------

364 -----

365 -----WAIT EXECUTION 2.8ms-----

366 -----

367 26		0x40		Read CONTINUE FLAG
--------	--	------	--	--------------------

368 -----

369 27			0x82	Read Status Code (82 = PRINT STRING & RET KEY)
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370 -----

371 28			0x01	Read Status Code (01 = No Error)
--------	--	--	------	----------------------------------

372 -----

373 29			0x41	Character ASCII to Print
374 30			0x42	Character ASCII to Print
375 31			0x43	Character ASCII to Print
376 32			0x44	Character ASCII to Print
377 33			0x45	Character ASCII to Print
378 34			0x00	End of String 00 (NULL)

379 -----

380 35			0x41	Send Character ASCII
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381 -----

382
383 PRINT CHARACTER (083H)

384 -----

385 THE PRINT CHARACTER FUNCTION (0X83H) MUST BE CALLED ONLY AT THE LAST STATUS CODE

386 OF THE COMMAND.

387 THE NEXT VALUE TO SEND IS THE ERROR CODE THAT SHOULD BE SENT BACK TO THE OPERATING

388 SYSTEM, ACCORDING TO THE TABLE OF VALID RETURN CODES. (Status Code 01 = No Error).

389 THE CHARACTER IS PRINTED ON THE MAIN SCREEN.

390 THEN THE ERROR CODE SHOULD BE SENT BACK TO THE OPERATING SYSTEM ACCORDING TO THE

391 TABLE OF RETURN CODES.

392 =====

394 BYTE	STAT	DATA	DATA	DESCRIPTION
395 NUMB	IN	OUT	IN	
396 ----	----	----	----	-----
397 PORT	RD-15	WR-14	RD-14	Z80 I/O PORT

398 =====

399 .

400 -----

401 25		0x20		Send CRC FROM TIMEX
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402 -----

403 -----WAIT EXECUTION 2.8ms-----

404 -----

405 26		0x40		Read CONTINUE FLAG
--------	--	------	--	--------------------

406 -----

407 27			0x83	Read Status Code (83 = PRINT CHARACTER)
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408 -----

409 28			0x01	Read Status Code (01 = No Error)
--------	--	--	------	----------------------------------

410 -----

411 29			0x41	Character ASCII to Print
412 30			0x42	Character ASCII to Print
413 31			0x43	Character ASCII to Print
414 32			0x44	Character ASCII to Print
415 33			0x45	Character ASCII to Print
416 34			0x00	End of String 00 (NULL)

417 -----

418 35			0x41	Send Character ASCII
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421
422 FUNCTION RETURN KEY (084H)
423 -----

424
425 THE RETURN KEY FUNCTION (0X84H) MUST BE CALLED ONLY AT THE LAST STATUS CODE OF
426 THE COMMAND.

427
428 THE NEXT VALUE TO SEND IS THE ERROR CODE THAT SHOULD BE SENT BACK TO THE OPERATING
429 SYSTEM, ACCORDING TO THE TABLE OF VALID RETURN CODES. (Status Code 01 = No Error).

430
431 THE SYSTEM WAITS TILL A KEY IS PRESSED, AND THE ASCII CODE OF THE KEY IS SENT TO
432 THE TS-PICO.

433
434 THEN THE ERROR CODE SHOULD BE SENT BACK TO THE OPERATING SYSTEM ACCORDING TO THE
435 TABLE OF RETURN CODES.

436
437 =====

438 BYTE	STAT	DATA	DATA	DESCRIPTION
439 NUMB	IN	OUT	IN	
440 ----	----	----	----	-----
441 PORT	RD-15	WR-14	RD-14	Z80 I/O PORT
=====				
443 .				
444 .	<<<<<<<< LAST PART OF 1ST MESSAGE BLOCK >>>>>>>>			
445 .				

447 25		0x20		Send CRC FROM TIMEX

-----WAIT EXECUTION 2.8ms-----				

451 26		0x40		Read CONTINUE FLAG

453 .				
454 .	<<<<<<<< END MESSAGE PHASE >>>>>>>>			
455 .				

457 27		0x84		Read Status Code (84 = RETURN KEY)

459 28		0x01		Read Exit Status Code (01 = No Error)
-----WAIT FOR A KEY INPUT -----				

-----WAIT EXECUTION 2.8ms-----				

464 29		0x40		Read CONTINUE FLAG

466 30		0x41		Read KEY Value "A"

491
492 FUNCTION GET STATUS (085H)
493 -----

494 THE GET STATUS FUNCTION (0X85H) MUST BE CALLED ONLY AT THE LAST STATUS CODE OF
495 THE COMMAND.

496
497 THE NEXT VALUE TO SEND IS THE ERROR CODE THAT SHOULD BE SENT BACK TO THE OPERATING
498 SYSTEM, ACCORDING TO THE TABLE OF VALID RETURN CODES. (Status Code 01 = No Error).
499

500 THE GET STATUS RETURNS A BYTE THAT CORRESPONDS TO 1 BIT PER DEVICE

501
502 B0 = KEYBOARD STATUS
503 B1 = AUX STATUS
504 B2 = PRINTER STATUS
505 B3 = DISK STATUS
506

507 AT THE CURRENT VERSION OF THE TS-PICO SOME OF THESE BITS MAY NOT BE SUPPORTED.
508 0= READY
509 1= BUSY
510

511 =====
512 BYTE STAT DATA DATA DESCRIPTION
513 NUMB IN OUT IN
514 -----
515 PORT RD-15 WR-14 RD-14 Z80 I/O PORT
516 =====

517 .
518 .
519 . <<<<<<<< LAST PART OF 1ST MESSAGE BLOCK >>>>>>>>
520 .

521 -----
522 25 0x20 Send CRC FROM TIMEX
523 -----

524 -----WAIT EXECUTION 2.8ms-----
525 -----

526 26 0x40 Read CONTINUE FLAG
527 -----

528 .
529 . <<<<<<<< END MESSAGE PHASE >>>>>>>>
530 .

531 -----
532 27 0x85 Read Status Code (85 = RETURN KEY)
533 -----

534 28 0x01 Read Exit Status Code (01 = No Error)
535 -----

536 29 0x41 Read STATUS Value
537 -----

538
539 FUNCTION PRINT STRING WITH LOOP (Y/N) (086H)
540 -----

541 THE PRINT_STRING WITH LOOP FUNCTION (0x86) MUST BE CALLED ONLY AT THE LAST STATUS
542 CODE OF THE COMMAND.

543
544 THE NEXT VALUE TO SEND IS THE ERROR CODE THAT SHOULD BE SENT BACK TO THE OPERATING
545 SYSTEM, ACCORDING TO THE TABLE OF VALID RETURN CODES. (Status Code 01 = No Error).
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562 THE STRING TO PRINT ON THE MAIN SCREEN WITH LOOP IS SENT CHARACTER BY CHARACTER
563 FINISHING WITH THE NULL CHARACTER. THE MESSAGE TO PRINT MUST END WITH THE LINE
564 "CONTINUE (Y/N)". THIS OPERATION IS REPEATED WHILE THE "N" KEY IS NOT PRESSED.
565

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=====
566
567 BYTE      STAT      DATA      DATA      DESCRIPTION
568 NUMB      IN        OUT        IN
569 ----      -
570 PORT      RD-15     WR-14     RD-14     Z80 I/O PORT
571 =====
572 .
573 .
574 -----
575 25          0x20          Send CRC FROM TIMEX
576 -----
577 -----WAIT EXECUTION 2.8ms-----
578 -----
579 26          0x40          Read CONTINUE FLAG
580 -----
581 27          0x86          Read Status Code (86 = PRINT STRING WITH LOOP)
582 -----
583 28          0x01          Return Status Code (01 = No Error)
584 -----
585 29          0x41          Character ASCII to Print
586 30          0x42          Character ASCII to Print
587 31          0x43          Character ASCII to Print
588 32          0x44          Character ASCII to Print
589 33          0x45          Character ASCII to Print
590 34          0x00          End of String 00 (NULL)
591 -----
592 -----WAIT FOR A KEY INPUT -----
593 -----
594 -----WAIT EXECUTION 2.8ms-----
595 -----
596 35          0x40          Read CONTINUE FLAG
597 -----
598 36          0x41          Read KEY Value "Y" or "N" ***
599 -----
600          ***          IF THE REPONSE IS "N" ENDS WITH Return Status Code
601                      ELSE GOTO STEP 29
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632 MESSAGES SUPPORTED BY VERSION 1.0 OF THE TS-2068 ROM VERSION 2023 40 ANNIVERSARY.
633 =====
634

635 SAVE "9876543210" CODE 0,2 (00 = SAVE)
636

637 CO RD-15 WR-14 RD-14
638 =====

639	00	0x00	BLOCK TYPE (00 = Header)
640	01	0x00	TADDR (00 = SAVE)
641	02	0xFF	BANK # (FF=HOME)
642	03	0x63	SESSION ID (Low Byte)
643	04	0x10	SESSION ID (High Byte)
644	05	0xA4	MEMORY ADDRESS (Low Byte)
645	06	0x6C	MEMORY ADDRESS (High Byte)
646	07	0x11	BLOCK LEN (Low Byte)
647	08	0x00	BLOCK LEN (High Byte)

648 -----

649 09 0x55 CRC FROM TIMEX
650 -----

651 10 0x01 Status Code (01 = No Error)
652 -----

653 11 0x40 -----WAIT EXECUTION-----
654 -----

655 12 0x00 BLOCK TYPE (00 = Header)

656 13 0x63 SESSION ID (Low Byte)

657 14 0x10 SESSION ID (High Byte)
658 -----

659 15 0x03 HDTYPE

660 16 0x39 "9"

661 17 0x38 "8"

662 18 0x37 "7"

663 19 0x36 "6"

664 20 0x35 "5"

665 21 0x34 "4"

666 22 0x33 "3"

667 23 0x32 "2"

668 24 0x31 "1"

669 25 0x30 "0"

670 26 0x02 BLOCK LEN (Low Byte)

671 27 0x00 BLOCK LEN (High Byte)

672 28 0x00 MEMORY ADDRESS (Low Byte)

673 29 0x00 MEMORY ADDRESS (High Byte)

674 30 0x75 HDVARS (Low Byte)

675 31 0x80 HDVARS (High Byte)
676 -----

677 32 0x86 CRC FROM TIMEX
678 -----

679 33 0x40 -----WAIT EXECUTION-----
680 -----

681 34 0x01 Status Code (01 = No Error)
682 -----

683 35 0xFF Block Type (FF = BLOCK DATA)

684 36 0x63 SESSION ID (Low Byte)

685 37 0x10 SESSION ID (High Byte)

686 38 0xF3 DATA(0000)

687 39 0x18 DATA(0001)
688 -----

689 40 0x67 CRC FROM TIMEX
690 -----

691 41 0x40 -----WAIT EXECUTION-----
692 -----

693 42 0x01 Status Code (01 = No Error)
694 -----
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```
701
702 LOAD "9876543210" CODE (01 = LOAD)
703 (02 = VERIFY)
704 CO RD-15 WR-14 RD-14 (03 = MERGE)
705 =====
706 00 0x00 BLOCK TYPE (00 = Header)
707 01 0x01 TADDR (01 = LOAD)
708 02 0xFF BANK # (FF=HOME)
709 03 0x6A SESSION ID (Low Byte)
710 04 0x86 SESSION ID (High Byte)
711 05 0xB6 MEMORY ADDRESS (Low Byte)
712 06 0x6C MEMORY ADDRESS (High Byte)
713 07 0x11 BLOCK LEN (Low Byte)
714 08 0x00 BLOCK LEN (High Byte)
715 -----
716 09 0xD9 CRC FROM TIMEX
717 -----
718 10 0x01 Status Code (01 = No Error)
719 -----WAIT EXECUTION-----
720 11 0x40 CONTINUE FLAG
721 -----
722 12 0x00 REQUEST BLOCK TYPE ID (00 = Header)
723 -----
724 13 0x00 BLOCK TYPE FROM FILE (00 = Header)
725 14 0x03 HDTYPE
726 15 0x39 "9"
727 16 0x38 "8"
728 17 0x37 "7"
729 18 0x36 "6"
730 19 0x35 "5"
731 20 0x34 "4"
732 21 0x33 "3"
733 22 0x32 "2"
734 23 0x31 "1"
735 24 0x30 "0"
736 25 0x02 BLOCK LEN (Low Byte)
737 26 0x00 BLOCK LEN (High Byte)
738 27 0x00 MEMORY ADDRESS (Low Byte)
739 28 0x00 MEMORY ADDRESS (High Byte)
740 29 0x00 HDVARS (Low Byte)
741 30 0x80 HDVARS (High Byte)
742 -----
743 31 0x80 CRC FROM FILE
744 -----
745 32 0x00 CRC FROM TIMEX
746 -----WAIT EXECUTION-----
747 33 0x40 CONTINUE FLAG
748 -----
749 34 0x01 Status Code (01 = No Error)
750 -----
```

```
751
752 <<<<<< LOAD "9876543210" CODE CONTINUE NEXT PAGE >>>>>>
753
754
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771
772 <<<<<< LOAD "9876543210" CODE CONTINUE >>>>>>
773

```

774 CO RD-15      WR-14      RD-14      (03 = MERGE)
775 -----PROCESS RETURN CMD-----
776 35            0xFF            BLOCK TYPE ID (FF = BLOCK DATA)
777 36            0x01            TADDR (00 = SAVE)
778 37            0xFF            BANK # (FF=HOME)
779 38            0x6A            SESSION ID (Low Byte)
780 39            0x86            SESSION ID (High Byte)
781 40            0x00            MEMORY ADDRESS (Low Byte)
782 41            0x00            MEMORY ADDRESS (High Byte)
783 42            0x02            BLOCK LEN (Low Byte)
784 43            0x00            BLOCK LEN (High Byte)
785 -----
786 44            0xEF            CRC FROM TIMEX
787 -----
788 45            0x01            Status Code (01 = No Error)
789 -----WAIT EXECUTION-----
790 46  0x40            CONTINUE FLAG
791 -----
792 47            0xFF            REQUEST BLOCK TYPE ID (FF = BLOCK DATA)
793 -----
794 48            0xFF            BLOCK TYPE FROM FILE (FF = BLOCK DATA)
795 49            0xF3            DATA(0000)
796 50            0x18            DATA(0001)
797 51            0x14            CRC FROM FILE
798 -----
799 52            0x00            CRC FROM TIMEX
800 -----WAIT EXECUTION-----
801 53  0x40            CONTINUE FLAG
802 -----
803 54            0x01            Status Code (01 = No Error)
804 -----

```

805
806
807 *****

```

808
809 ; *****
810 ; ERROR CODE RETURN ON BLOCK          * TADDR HEADERS VALUES*
811 ; *****
812 0  = REPORT J - INVALID I/O DEVICE    00 = SAVE
813 1  = OK                               01 = LOAD
814 2  = REPORT R - TAPE LOADING ERROR    02 = VERIFY
815 3  = REPORT F - INVALID FILE NAME     03 = MERGE
816 4  = REPORT Q - PARAMETER ERROR       04 = COPY
817 5  = REPORT C - NON SENSE IN BASIC    05 = LPRINT
818 6  = REPORT 6 - NUMBER TOO BIG        06 = NEW      * NOT USED IN PICO *
819 7  = REPORT 8 - END OF FILE           07 = RUN      * NOT USED IN PICO*
820 8  = REPORT A - INVALID ARGUMENT
821 9  = REPORT 9 - STOP
822 10 = REPORT J - INVALID I/O DEVICE
823 +10 = REPORT D - BREAK-CONT REPEATS
824

```

```

825 *****
826 * BLOCK TYPE *      * RET CODE FUNCTIONS      *      * FRAME ID's *
827 *****
828 00 = HEADER         081H      PRINT STRNG      "A" (ASM COMND)
829 FF = DATA BLOC    082H      PRINT STRNG & RET KEY "B" (BASIC COMND)
830                     083H      PRINT CHARACTER    "C" (CP/M CBIOS COMND)
831                     084H      RETURN KEY         "D" (DATA BLOCK)
832                     085H      GET STATUS         "E" (EXTRA DATA)
833                     086H      PRINT LOOP STRNG Y/N
834
835
836

```

837 *****

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```
840 TS2068-PICO TPI CMD SPECS AGO 18 2023 V:2.4 MACRO-80 3.44 09-Dec-81 PAGE 1-13
841
842 COPY (04 = COPY SCREEN) SAVE "TPI:DELETE" CODE 4386,13124 (00 = SAVE CMD)
843 -----
844 CO RD-15 WR-14 RD-14 CO RD-15 WR-14 RD-14
845 =====
846 00 0x42 "B" (BASIC COMMAND) 00 0x42 "B" (BASIC COMND)
847 01 0x04 COPY (TADDR) 01 0x00 SAVE (TADDR)
848 02 0xFF BANK NUMBER (FF=HOME) 02 0xFF BANK NUMBER (FF=HOME)
849 03 0xFF COLOR MODE (FF=ATTR) 03 0x22 PMR 1 L
850 04 0x00 VIDEO MODE 04 0x11 PMR 1 H
851 05 0x20 COLUMNS (X-MAX) 05 0x44 PMR 2 L
852 06 0x18 LINES (Y-MAX) 06 0x33 PMR 2 H
853 07 0x00 DATA LEN L 07 0x0A COMND LEN L
854 08 0x1B DATA LEN H 08 0x00 COMND LEN H
855 -----
856 09 0x65 CRC FROM TIMEX 09 0xF3 CRC FROM TIMEX
857 -----
858 11 0x01 Status Code 10 0x01 Status Code
859 -----WAIT EXECUTION-----WAIT EXECUTION-----
860 10 0x40 CONTINUE FLAG 11 0x40 CONTINUE FLAG
861 -----
862 12 0x44 "D" (DATA BLOCK CMD) 12 0x44 "D" (DATA BLOCK CMD)
863 13 0x00 COMND LEN L 13 0x0A COMND LEN L
864 14 0x1B COMND LEN H 14 0x00 COMND LEN H
865 15 0x00 SCREEN DATA 0 15 0x54 "T"
866 16 0x00 SCREEN DATA 1 16 0x50 "P"
867 17 0x00 . 17 0x49 "I"
868 . 0x00 . 18 0x3A ":"
869 . 0x00 . 19 0x44 "D"
870 . 0x00 . 20 0x45 "E"
871 6922 0x0F . 21 0x4C "L"
872 6923 0x0F . 22 0x45 "E"
873 6924 0x0F SCREEN DATA 6910 23 0x54 "T"
874 6925 0x0F SCREEN DATA 6911 24 0x45 "E"
875 -----
876 6926 0x93 CRC FROM TIMEX 25 0x20 CRC FROM TIMEX
877 -----WAIT EXECUTION-----WAIT EXECUTION-----
878 6927 0x40 CONTINUE FLAG 26 0x40 CONTINUE FLAG
879 -----
880 6928 0x01 Status Code 27 0x01 Status Code
881 -----
882
883 *****
884
885 LPRINT 1 (05 = PRINT ASCII CHAR) <<<< OPTIONAL MSG IF NOT ENDS WITH ";" >>>>
886 -----
887 CO RD-15 WR-14 RD-14 CO RD-15 WR-14 RD-14
888 =====
889 00 0x42 "B" (BASIC COMMAND) 00 0x42 "B" (BASIC COMMAND)
890 01 0x05 LPRINT CHR (TADDR) 01 0x05 LPRINT CHR (TADDR)
891 02 0xFF BANK NUMBER (FF=HOME) 02 0xFF BANK NUMBER (FF=HOME)
892 03 0x31 ASCII CHARACTER 03 0x0D ASCII CHARACTER
893 04 0x01 CHR TYPE 04 0x01 CHR TYPE
894 05 0x01 PRINT FLAGS 05 0x01 PRINT FLAGS
895 06 0x0F COLOR ATTRIB (ATTRP) 06 0x0F COLOR ATTRIB (ATTRP)
896 07 0x00 DATA LEN L 07 0x00 DATA LEN L
897 08 0x00 DATA LEN H 08 0x00 DATA LEN H
898 -----
899 09 0x86 CRC FROM TIMEX 09 0xBA CRC FROM TIMEX
900 -----
901 11 0x01 Status Code 11 0x01 Status Code
902 -----WAIT EXECUTION-----WAIT EXECUTION-----
903 10 0x40 CONTINUE FLAG 10 0x40 CONTINUE FLAG
904 -----
905
906
907 CHR TYPE = 01=ASCII LEN 00 = NO DATA BLOCK PRINT FLAGS BIT 0,1 LISTING
908 CHR TYPE = 02=BITMAP-UDG LEN 08 = BITMAP CHAR DATA PRINT FLAGS BIT 4 AUTO
909
910
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```

```
911
912 LPRINT CHR$ 138;          ( BITMAP CHARACTER)
913
914 -----
915 CO RD-15    WR-14    RD-14
916 =====
917 00          0x42      "B" (BASIC COMMAND)
918 01          0x05      LPRINT CHR (TADDR)
919 02          0xFF      BANK NUMBER (FF=HOME)
920 03          0x8A      ASCII CHARACTER
921 04          0x02      CHR TYPE
922 05          0x01      PRINT FLAGS
923 06          0x0F      COLOR ATTRIBUTES (ATTRP)
924 07          0x08      DATA LEN L
925 08          0x00      DATA LEN H
926 -----
927 09          0x36      CRC FROM TIMEX
928 -----
929 10          0x01      Status Code
930 -----WAIT EXECUTION-----
931 11 0x40          CONTINUE FLAG
932 -----
933
934 <<<<<<  OPTIONAL MESSAGE PART  >>>>>>
935 <<<<<<  FOR BITMAP CHAR ONLY   >>>>>>
936 -----
937 12          0x44      "D" (DATA BLOCK CMD)
938 13          0x08      COMND LEN L
939 14          0x00      COMND LEN H
940 15          0xF0      "*****"
941 16          0xF0      "*****"
942 17          0xF0      "*****"
943 18          0xF0      "*****"
944 19          0xF0      "*****"
945 20          0xF0      "*****"
946 21          0xF0      "*****"
947 22          0xF0      "*****"
948 -----
949 23          0x4C      CRC FROM TIMEX
950 -----WAIT EXECUTION-----
951 24 0x40          CONTINUE FLAG
952 -----
953 25          0x01      Status Code
954 -----PROCESS RESPONSE-----
955
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980
```

981
982 LPRINT CHR\$ 144; (UDG CHARACTER "A")
983
984 -----

985 CO RD-15 WR-14 RD-14
986 =====

987 00 0x42 "B" (BASIC COMMAND)
988 01 0x05 LPRINT CHR (TADDR)
989 02 0xFF BANK NUMBER (FF=HOME)
990 03 0x90 ASCII CHARACTER
991 04 0x02 CHR TYPE
992 05 0x01 PRINT FLAGS
993 06 0x0F COLOR ATTRIBUTES (ATTRP)
994 07 0x08 DATA LEN L
995 08 0x00 DATA LEN H
996 -----

997 09 0x2C CRC FROM TIMEX
998 -----

999 10 0x01 Status Code
1000 -----WAIT EXECUTION-----

1001 11 0x40 CONTINUE FLAG
1002 -----

1003
1004 <<<<<< OPTIONAL MESSAGE PART >>>>>>
1005 <<<<<< FOR BITMAP CHAR ONLY >>>>>>
1006 -----

1007 12 0x44 "D" (DATA BLOCK CMD)
1008 13 0x08 COMND LEN L
1009 14 0x00 COMND LEN H
1010 15 0x00 " "
1011 16 0x3C " ***** "
1012 17 0x42 " * * "
1013 18 0x42 " * * "
1014 19 0x7E " * * * * * "
1015 20 0x42 " * * "
1016 21 0x42 " * * "
1017 22 0x00 " "
1018 -----

1019 23 0x0E CRC FROM TIMEX
1020 -----WAIT EXECUTION-----

1021 24 0x40 CONTINUE FLAG
1022 -----

1023 25 0x01 Status Code
1024 -----PROCESS RESPONSE-----

1025
1026
1027 SAVE "TPI:BMPSIZE" CODE 256,192 BMP RESOLUTION STD HIGH RESOLUTION
1028 256 x 192 512 x 192
1029 512 x 384 1024 x 384
1030 1024 x 768 2048 x 768
1031 2048 x 1596 4096 x 1596

1032 SAVE "TPI:TCAPST" START CAPTURE IN TEXT FILE

1033 SAVE "TPI:TCAPEN" END CAPTURE IN TEXT FILE
1034

1035 SAVE "TPI:TXTLIN" CODE 72 SET 72 LINES PER PAGE

1036 SAVE "TPI:TXTCOL" CODE 132 SET 132 CHARACTER PER LINE

1037 SAVE "TPI:TXTFON" CODE XXX SET FONT XXX

1038 SAVE "TPI:TXTEOP" SEND END OF PAGE
1039
1040
1041
1042
1043
1044
1045
1046
1047
1048
1049
1050
1050
1050
1050
1050
1050

1051
1052 LPRINT "1"; INK 4;"1"; PAPER 7;"3"
1053

1054 -----
1055 CO RD-15 WR-14 RD-14 01 02 03 04
1056 =====
1057 00 0x42 "B" (BASIC COMMAND) 0x42 0x42 0x42 0x42
1058 01 0x05 LPRINT CHR (TADDR) 0x05 0x05 0x05 0x05
1059 02 0xFF BANK NUMBER (FF=HOME) 0xFF 0xFF 0xFF 0xFF
1060 03 0x31 ASCII CHARACTER 0x10 0x10 0x04 0x32
1061 04 0x01 CHR TYPE 0x01 0x01 0x01 0x01
1062 05 0x01 PRINT FLAGS 0x01 0x01 0x01 0x01
1063 06 0x0F COLOR ATTRIBUTES 0x0F 0x0F 0x0F 0x0F
1064 07 0x00 DATA LEN L 0x00 0x00 0x00 0x00
1065 08 0x00 DATA LEN H 0x00 0x00 0x00 0x00
1066 -----
1067 09 0x86 CRC FROM TIMEX 0xA7 0xA7 0xB3 0x85
1068 -----
1069 10 0x01 Status Code 0x01 0x01 0x01 0x01
1070 -----WAIT EXECUTION-----
1071 11 0x40 CONTINUE FLAG 0x40 0x40 0x40 0x40
1072 -----

1073 <<<<<< CONTINUE SEQUENCE >>>>>>
1074 -----

1075 CO RD-15 WR-14 RD-14 05 06 07 08
1076 =====
1077 00 0x42 "B" (BASIC COMMAND) 0x42 0x42 0x42 0x42
1078 01 0x05 LPRINT CHR (TADDR) 0x05 0x05 0x05 0x05
1079 02 0xFF BANK NUMBER (FF=HOME) 0xFF 0xFF 0xFF 0xFF
1080 03 0x31 ASCII CHARACTER 0x11 0x07 0x33 0x0D
1081 04 0x01 CHR TYPE 0x01 0x01 0x01 0x01
1082 05 0x01 PRINT FLAGS 0x01 0x01 0x01 0x01
1083 06 0x0F COLOR ATTRIBUTES 0x0F 0x0F 0x0F 0x0F
1084 07 0x00 DATA LEN L 0x00 0x00 0x00 0x00
1085 08 0x00 DATA LEN H 0x00 0x00 0x00 0x00
1086 -----
1087 09 0x86 CRC FROM TIMEX 0xA6 0xB0 0x84 0xBA
1088 -----
1089 10 0x01 Status Code 0x01 0x01 0x01 0x01
1090 -----WAIT EXECUTION-----
1091 11 0x40 CONTINUE FLAG 0x40 0x40 0x40 0x40
1092 -----
1093

1094 FUNC BASIC
1095 CODE INSTRUCTION FUNCTION
1096 -----
1097 10H = INK (SELECT INK COLOR)
1098 11H = PAPER (SELECT PAPER COLOR)
1099 12H = FLASH (SELECT UNDERLINE CONTROL)
1100 13H = BRIGHT (SELECT BOLD LETER CONTROL)
1101 14H = INVERSE (SELECT ITALIC)
1102 15H = OVER xx (SEND ASCII EXTEND TO PRINTER (128 to 255) CHR\$ (128 + xx)
1103 -----
1104
1105
1106
1107
1108
1109
1110
1111
1112
1113
1114
1115
1116
1117
1118
1119
1120
1120
1120
1120
1120
1120

1121
 1122 LPRINT OVER 37 REM SEND Ñ TO PRINTER + <ENTER>
 1123 LPRINT OVER 37; REM SEND Ñ TO PRINTER

1124
 1125 Envia caracter ascii (128+37) = 165

CO RD-15	WR-14	RD-14	01	02
00	0x42	"B" (BASIC COMMAND)	0x42	0x42
01	0x05	LPRINT CHR (TADDR)	0x05	0x05
02	0xFF	BANK NUMBER (FF=HOME)	0xFF	0xFF
03	0x15	ASCII CHARACTER	0x25	0x0D
04	0x01	CHR TYPE	0x01	0x01
05	0x01	PRINT FLAGS	0x01	0x01
06	0x0F	COLOR ATTRIBUTES	0x0F	0x0F
07	0x00	DATA LEN L	0x00	0x00
08	0x00	DATA LEN H	0x00	0x00
09	0xA2	CRC FROM TIMEX	0x92	0xBA
10	0x01	Status Code	0x01	0x01
--WAIT EXECUTION--				
11 0x40		CONTINUE FLAG	0x40	0x40

1146 LPRINT AT 10,20;"A" REM LOCATE PRINTER HEAD OVER PAPER

CO RD-15	WR-14	RD-14	01	02	03	04
00	0x42	"B" (BASIC COMMAND)	0x42	0x42	0x42	0x42
01	0x05	LPRINT CHR (TADDR)	0x05	0x05	0x05	0x05
02	0xFF	BANK NUMBER (FF=HOME)	0xFF	0xFF	0xFF	0xFF
03	0x16	ASCII CHARACTER	0x0A	0x14	0x41	0x0D
04	0x01	CHR TYPE	0x01	0x01	0x01	0x01
05	0x01	PRINT FLAGS	0x01	0x01	0x01	0x01
06	0x0F	COLOR ATTRIBUTES	0x0F	0x0F	0x0F	0x0F
07	0x00	DATA LEN L	0x00	0x00	0x00	0x00
08	0x00	DATA LEN H	0x00	0x00	0x00	0x00
09	0xA1	CRC FROM TIMEX	0xBD	0xA3	0xF6	0xBA
10	0x01	Status Code	0x01	0x01	0x01	0x01
--WAIT EXECUTION--						
11 0x40		CONTINUE FLAG	0x40	0x40	0x40	0x40

1168 LPRINT AT 10,20,"A" REM LOCATE PRINTER HEAD OVER PAPER

CO RD-15	WR-14	RD-14	01	02	01	03	04
00	0x42	"B" (BASIC COMMAND)	0x42	0x42	0x42	0x42	0x42
01	0x05	LPRINT CHR (TADDR)	0x05	0x05	0x05	0x05	0x05
02	0xFF	BANK NUMBER (FF=HOME)	0xFF	0xFF	0xFF	0xFF	0xFF
03	0x16	ASCII CHARACTER	0x0A	0x14	0x06	0x41	0x0D
04	0x01	CHR TYPE	0x01	0x01	0x01	0x01	0x01
05	0x01	PRINT FLAGS	0x01	0x01	0x01	0x01	0x01
06	0x0F	COLOR ATTRIBUTES	0x0F	0x0F	0x0F	0x0F	0x0F
07	0x00	DATA LEN L	0x00	0x00	0x00	0x00	0x00
08	0x00	DATA LEN H	0x00	0x00	0x00	0x00	0x00
09	0xA1	CRC FROM TIMEX	0xBD	0xA3	0xB1	0xF6	0xBA
10	0x01	Status Code	0x01	0x01	0x01	0x01	0x01
--WAIT EXECUTION--							
11 0x40		CONTINUE FLAG	0x40	0x40	0x40	0x40	0x40

1191
1192 SAVE "TPI:BMPSIZE" CODE 256,192 (00 = SAVE CMD) SAVE "TPI:TXTSIZE" CODE 132,72 (00 = SAVE CMD)
1193
1194

1195 CO RD-15 WR-14 RD-14
1196 =====
1197 00 0x42 "B" (BASIC COMND)
1198 01 0x00 SAVE COMMAND (TADDR)
1199 02 0xFF BANK NUMBER (FF=HOME)
1200 03 0x00 PMR 1 L
1201 04 0x01 PMR 1 H
1202 05 0xC0 PMR 2 L
1203 06 0x00 PMR 2 H
1204 07 0x0B COMND LEN L
1205 08 0x00 COMND LEN H
1206 -----

1207 09 0x77 CRC FROM TIMEX
1208 -----

1209 10 0x01 Status Code
1210 -----WAIT EXECUTION-----

1211 11 0x40 CONTINUE FLAG
1212 -----

1213 12 0x44 "D" (DATA BLOCK CMD)
1214 13 0x0B COMND LEN L
1215 14 0x00 COMND LEN H
1216 15 0x54 "T"
1217 16 0x50 "P"
1218 17 0x49 "I"
1219 18 0x3A ":"
1220 19 0x42 "B"
1221 20 0x4D "M"
1222 21 0x50 "P"
1223 22 0x53 "S"
1224 23 0x49 "I"
1225 24 0x5A "Z"
1226 25 0x45 "E"
1227 -----

1228 26 0x62 CRC FROM TIMEX
1229 -----WAIT EXECUTION-----

1230 27 0x40 CONTINUE FLAG
1231 -----

1232 28 0x01 Status Code
1233 -----
1234
1235
1236
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1260

CO RD-15 WR-14 RD-14
=====
00 0x42 "B" (BASIC COMND)
01 0x00 SAVE COMMAND (TADDR)
02 0xFF BANK NUMBER (FF=HOME)
03 0x84 PMR 1 L
04 0x00 PMR 1 H
05 0x48 PMR 2 L
06 0x00 PMR 2 H
07 0x0B COMND LEN L
08 0x00 COMND LEN H

09 0x7A CRC FROM TIMEX

10 0x01 Status Code
-----WAIT EXECUTION-----

11 0x40 CONTINUE FLAG

12 0x44 "D" (DATA BLOCK CMD)
13 0x0B COMND LEN L
14 0x00 COMND LEN H
15 0x54 "T"
16 0x50 "P"
17 0x49 "I"
18 0x3A ":"
19 0x42 "T"
20 0x4D "X"
21 0x50 "T"
22 0x53 "S"
23 0x49 "I"
24 0x5A "Z"
25 0x45 "E"

26 0x62 CRC FROM TIMEX
-----WAIT EXECUTION-----

27 0x40 CONTINUE FLAG

28 0x01 Status Code

1260

```
1260 TS2068-PICO TPI CMD SPECS AGO 18 2023 V:2.4 MACRO-80 3.44 09-Dec-81 PAGE 1-19
1261
1262 ;-----
1263 ; SAVE "TPI:MEMBOOT" CODE 1,2 SET ACTIVE MEMORY FOR HOME & EXROM (SET PAGE 2)
1264 ;
1265 ; SAVE "TPI:MEMBOOT" CODE 1,0 SET ACTIVE MEMORY FOR HOME & EXROM (PAGE DEFAULT IN CONFIG FILE)
1266 ;-----
1267 ;
1268 ; PMR1: SET ( 0 = DEFAULT / 1 = ROM / 2 = RAM / 3 = NONE)
1269 ; PMR2: PAGE #(A16-A18)
1270 ;-----
1271 ; 00 42 "B"
1272 ; 01 00 SAVE COMMAND (TADDR)
1273 ; 02 FF BANK # (00=DOCK FF=HOME ...)
1274 ; 03 01 PMR1 SET MEMORY LOCATION (1 = ROM / 2 = RAM)
1275 ; 04 00
1276 ; 05 00 PMR2 PAGE #(A16-A18)
1277 ; 06 00
1278 ; 07 0B BLK_LEN
1279 ; 08 00
1280 ;-----
1281 ; 09 B7 CRC
1282 ;-----
1283 ; 10 01 STATUS CODE
1284 ;-----
1285 ; 11 40 WAIT CONTINUE
1286 ;-----
1287 ; 12 44 "D"
1288 ; 13 0B BLK_LEN
1289 ; 14 00
1290 ;-----
1291 ; 15 54 "T"
1292 ; 16 50 "P"
1293 ; 17 49 "I"
1294 ; 18 3A ":"
1295 ; 19 4D "M"
1296 ; 20 45 "E"
1297 ; 21 4D "M"
1298 ; 22 42 "B"
1299 ; 23 4F "O"
1300 ; 24 4F "O"
1301 ; 25 54 "T"
1302 ; 26 6B CRC
1303 ;-----
1304 ; 27 40 WAIT CONTINUE
1305 ;-----
1306 ; 28 01 STATUS CODE
1307 ;-----
1308
1309
1310 ;*****
1311
1312
1313
1314
1315
1316
1317
1318
1319
1320
1321
1322
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1324
1325
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```

```
1331 ;-----
1332 ;
1333 ; SAVE "TPI:READY"
1334 ;-----
1335 ; 00          42          "B"
1336 ; 01          00          SAVE COMMAND (TADDR)
1337 ; 02          FF          BANK # (00=DOCK FF=HOME ...)
1338 ; 03          00          PMR1      (NOT USED)
1339 ; 04          00
1340 ; 05          00          PMR2      (NOT USED)
1341 ; 06          00
1342 ; 07          09          BLK_LEN
1343 ; 08          00
1344 ;-----
1345 ; 09          B4          CRC
1346 ;-----
1347 ; 10          01          STATUS CODE
1348 ;-----
1349 ; 11          40          WAIT CONTINUE
1350 ;-----
1351 ; 12          44          "D"
1352 ; 13          09          BLK_LEN
1353 ; 14          00
1354 ;-----
1355 ; 15          54          "T"
1356 ; 16          50          "P"
1357 ; 17          49          "I"
1358 ; 18          3A          ":"
1359 ; 19          52          "R"
1360 ; 20          45          "E"
1361 ; 21          41          "A"
1362 ; 22          44          "D"
1363 ; 23          59          "Y"
1364 ;-----
1365 ; 24          71          CRC
1366 ;-----
1367 ; 25          40          WAIT CONTINUE
1368 ;-----
1369 ; 26          01          STATUS CODE
1370 ;-----
1371 ;
1372 ; *****
1373 ;
1374 ;
1375 ;
1376 ;
1377 ;
1378 ;
1379 ;
1380 ;
1381 ;
1382 ;
1383 ;
1384 ;
1385 ;
1386 ;
1387 ;
1388 ;
1389 ;
1390 ;
1391 ;
1392 ;
1393 ;
1394 ;
1395 ;
1396 ;
1397 ;
1398 ;
1399 ;
1400 ;
1400 ;
1400 ;
1400 ;
1400 ;
1400 ;
```



```

1400 TS2068-PICO TPI CMD SPECS   AGO 18 2023 V:2.4   MACRO-80 3.44   09-Dec-81 PAGE 1-21
1401
1402 ;-----
1403 ; COPY TO DOCK RAM PAGE DATA AND SET AS DEFAULT CARTRIDGE BOOT
1404 ;
1405 ; SAVE "TPI:LDDOCK" CODE P_INFO,PAGE   SAVE "TPI:LDDOCK" CODE 13,2
1406 ; SAVE "TPI:LDDOCK" CODE MODE,PAGE/SUB_BANK
1407 ; SAVE "TPI:LDDOCK"           (NO PARAM, LOAD DEFAULT IN RAM PAGE 0 FROM FILE 1st option to load
1408 ;                             TS2068.TCC OR 2ND load TS2068.DCK file from SD Card)
1409 ;
1410 ;                             (P_INFO Param)
1411 ; SAVE "TPI:LDDOCK" CODE 1,0   END METOD MODE = 0 LOAD AUTO IMAGE FROM ROM PAGE 08
1412 ;                             END METOD MODE = 1 NEW
1413 ;                             END METOD MODE = 2 RESET
1414 ;                             END METOD MODE = 3 JUMP TO LOAD_INIT_ADDR (Used in .TCC Format)
1415 ;                             END METOD MODE = 4 MEMORY FILL WITH 00
1416 ;                             END METOD MODE = 5 NO ACTION, ONLY COPY RAM FROM ROM
1417 ;
1418 ;                             ROM SOURCE PAGE SETTING TO COPY DOCK RAM
1419 ;                             END METOD MODE = 8 DOCK SOURCE IS ROM PAGE 8 BINARY ="001000"
1420 ;                             END METOD MODE = 16 DOCK SOURCE IS ROM PAGE 9 BINARY ="010000"
1421 ;                             END METOD MODE = 24 DOCK SOURCE IS ROM PAGE 10 BINARY ="011000"
1422 ;                             END METOD MODE = 32 DOCK SOURCE IS ROM PAGE 11 BINARY ="100000"
1423 ;                             END METOD MODE = 40 DOCK SOURCE IS ROM PAGE 12 BINARY ="101000"
1424 ;                             END METOD MODE = 48 DOCK SOURCE IS ROM PAGE 13 BINARY ="110000"
1425 ;                             END METOD MODE = 56 DOCK SOURCE IS ROM PAGE 14 BINARY ="111000"
1426 ;                             END METOD MODE + 80H (Bit 7 = 1) SET DOCK BANK BANK AS RAM R/O
1427 ;
1428 ;                             PAGE/SUB_BANK
1429 ;                             (PAGE (0-7) - OPTIONAL RAM PAGE TO COPY ROM (DEST.DEFAULT = 00))
1430 ;-----
1431 ; THE FILENAME OF DOCK IMAGEN IS LOCATED IN PARAMETER TABLE
1432 ;-----
1433 ; 00          42          "B"
1434 ; 01          00          SAVE COMMAND (TADDR)
1435 ; 02          FF          BANK (EXECUTION BANK NUMBER)
1436 ; 03          0D          PMR1 = END METOD MODE
1437 ; 04          00
1438 ; 05          02          PMR2 = (MEMORY PAGE SETUP/SUB_BANK NUMBER)
1439 ; 06          00
1440 ; 07          0A          COMMAND BLK_LEN
1441 ; 08          00
1442 ;-----
1443 ; 09          B8          CRC
1444 ;-----
1445 ; 10          01          STATUS CODE
1446 ;-----
1447 ; 11          40          WAIT CONTINUE
1448 ;-----
1449 ; 12          44          "D"
1450 ; 13          0A          COMMAND BLK_LEN
1451 ; 14          00
1452 ;-----
1453 ; 15          54          "T"
1454 ; 16          50          "P"
1455 ; 17          49          "I"
1456 ; 18          3A          ":"
1457 ; 19          4C          "L"
1458 ; 20          44          "D"
1459 ; 21          44          "D"
1460 ; 22          4F          "O"
1461 ; 23          43          "C"
1462 ; 24          4B          "K"
1463 ;-----
1464 ; 25          32          CRC
1465 ;-----
1466 ; 26          40          WAIT CONTINUE
1467 ;-----
1468 ; 27          01          STATUS CODE
1469 ;-----
1470
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1470 TS2068-PICO TPI CMD SPECS AGO 18 2023 V:2.4 MACRO-80 3.44 09-Dec-81 PAGE 1-22
1471
1472 ; *****
1473 ;
1474 ; -----
1475 ; COMMAND PARAMETERS TABLE
1476 ; -----
1477 ;
1478 ; -----
1479 ; PARAMETERS ID
1480 ; -----
1481 ; 00-1F ERROR CODES
1482 ; -----
1483 ; REQUEST PARAMETERS ID
1484 ; -----
1485 ; 20 BINARY BLOCK
1486 ; 21 ASCII NAME 1
1487 ; 22 ASCII NAME 2
1488 ; 23 ASCII NAME 3
1489 ; -----
1490 ; DATA PARAMETER BLOCK TYPE ID
1491 ; -----
1492 ; 27 = NO MORE PARAMETERS
1493 ; 28 = CARTRIDGE HEADER
1494 ; 29 = CARTRIDGE RAW IMAGEN LOW (CHUNK 0 TO 6)
1495 ; 2A = CARTRIDGE RAW IMAGEN HIGH (CHUNK 7)
1496 ; 2B = SNAPSHOT FILENAME
1497 ; 2C = SNAPSHOT HARDWARE REGISTERS
1498 ; 2D = SNAPSHOT HOME DATA
1499 ; 2E = SNAPSHOT DOCK DATA
1500 ; 2F = SNAPSHOT EXP.BANK DATA
1501 ; -----
1502 ; ASSEMBLER COMMAND TABLE
1503 ; -----
1504 ;
1505 ; COMMAND ID FUNCTION DESCRIPTION PARAMETER
1506 ; -----
1507 ; 040H STATUS CHECK CONECTIVITY / RETURN STATUS 20
1508 ; 041H GETMPR GET CURRENT MEMORY PAGE REGISTER VALUE (00 TO 0F) 20
1509 ; 042H SETMPR SET MEMORY PAGE REGISTER VALUE (00 TO 0F) 20
1510 ; 043H SETDMR SET DEFAULT MEMORY REGISTER (VALUE FROM CONFIG FILE) 20
1511 ; 044H CLOSPMT CLOSE PARAMETER BLOCK / COMMAND ENDS 20
1512 ; 045H LDDOCK LOAD CARTRIDGE IMAGE IN DOCK BANK 21,28,29,2A
1513 ; 046H TAPPOS MOVE TAP POINTER POSITION (1=BEG,2=END,3=NXT_HDR,4=NXT_BLK)
1514 ; 047H SNSNAP SET SNAPSHOT NAME
1515 ; 048H SASNAP SAVE SNAPSHOT IMAGE
1516 ; 049H LDSNAP LOAD SNAPSHOT IMAGE
1517 ;
1518 ; 050H DATARD RAW DATA READ
1519 ; 051H DATAWR RAW DATA WRITE
1520 ; 052H CTRLRD CONTROL DATA READ
1521 ; 053H CTRLWR CONTROL DATA WRITE
1522 ;
1523 ; -----
1524 ; ; NON BLOCKING ROUTINES
1525 ; -----
1526 ;
1527 ; 054H SNDNBC SEND NON BLOCKING COMMAND
1528 ; 055H POLING CHECK IF LAST COMMAND WAS EXECUTED
1529 ; 056H ABORT ABORT LAST SESSION ID COMMAND
1530 ; 057H BUSY CHECK IF THE COMMUNIC. IS BUSY
1531 ; -----
1532
1533 ; *****
1534 ;
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1541
1542 ;-----
1543 ;      SEND LDDOCK PARAM (INFORMATION LOCATED COMMAND PARAMETER AREA)
1544 ;      *****
1545 ;      ****          AVAILABLE ONLY FROM ASSEMBLER          ****
1546 ;      ****          TPIBIOS ROUTINE LDDOCK - 0E027H (57383)      ****
1547 ;      *****
1548 ;-----
1549
1550
1551 ;-----
1552 ; COMMAND LDDOCK
1553 ;-----
1554 ; 00          44          "A"          ; ASSEMBLER TYPE COMMAND
1555 ; 01          45          COMMAND ID (45=LDDOCK)
1556 ; 02          00          BANK # (00=DOCK FF=HOME ...)
1557 ; 03          00          SESION ID LOW
1558 ; 04          00          SESION ID HIGH
1559 ;-----
1560 ; 05          21          COMMAND PARAMETER ID ( 21= SINGLE ASCII PARAM) FILENAME
1561 ; 06          03          COMMAND BLOCK NUMBER (00 = LAST COMMAND BLOCK MESSAGE / 01..02..)
1562 ; 07          0C          BLK_LEN LOW
1563 ; 08          00          BLK_LEN HIGH
1564 ;-----
1565 ; 09          6B          CRC
1566 ;-----
1567 ; 10          40          WAIT CONTINUE
1568 ;-----
1569 ; 11          01          STATUS CODE
1570 ;-----
1571 ; 12          44          "D"
1572 ; 13          0C          BLK_LEN
1573 ; 14          00
1574 ;-----
1575 ;      DATA COMMAND ARGUMENT
1576 ;-----
1577 ; 15          46          "F"          ; Cartridge Filename on the SD
1578 ; 16          49          "I"
1579 ; 17          4C          "L"
1580 ; 18          45          "E"
1581 ; 19          4E          "N"
1582 ; 20          41          "A"
1583 ; 21          4D          "M"
1584 ; 22          45          "E"
1585 ; 23          2E          ". "      ". "
1586 ; 24          44          "D"      "T"
1587 ; 25          43          "C"      "C"
1588 ; 26          4B          "K"      "C"
1589 ;-----
1590 ; 27          6B          CRC
1591 ;-----
1592 ; 28          01          STATUS CODE
1593 ;-----
1594 ; 29          40          WAIT CONTINUE
1595 ;-----
1596 ; 30          28          REQUEST DATA PARAMETER BLOCK TYPE ID (28 = CARTRIDGE HEADER)
1597 ;                          (OR ERROR CODE IF VALUE <20H OR >80H)
1598 ;-----
1599
1600
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1611
1612 ;-----
1613 ; COMMAND LDDOCK CARTRIDGE INFO TABLE
1614 ;-----
1615 ; 31          44      "E"      RESPONSE TO COMMAND
1616 ; 32          1C      BLK_LEN
1617 ; 33          00
1618 ;-----
1619 ; 34          45      COMMAND ID (45=LDDOCK)
1620 ; 35          00      BANK # (00=DOCK FF=HOME ...)
1621 ; 36          00      SESION ID LOW
1622 ; 37          00      SESION ID HIGH
1623 ;-----
1624 ; 38          28      DATA PARAMETER BLOCK TYPE ID ( 28 = CARTRIDGE HEADER )
1625 ; 39          02      COMMAND BLOCK NUMBER (00 = SINGLE COMMAND BLOCK/LAST MESSAGE / N
1626 ; 40          00      LOADING ADDR LOW (0000 = N/A FOR CARTRIDGE HEADER)
1627 ; 41          00      LOADING ADDR HIGH
1628 ; 42          0B      CARTRIDGE HEADER LEN LOW
1629 ; 43          00      CARTRIDGE HEADER LEN HIGH
1630 ;-----
1631 ; 44          00      01=SINGLE DATA DOCK BLOCK (MAX 56k) OR NN=MULTIPLE DATA DOCK BLO
1632 ; 45          00      FLAG OF USE DATA IN CHUNK 7
1633 ; 46          00      DESTINATION BANK NUMBER
1634 ; 47          03      HS SETTINGS ON LOAD (ACTIVE IN 1 CHUNK 3 & 7 MUST BE DISABLED IN
1635 ; 48          01      CARTRIDGE TYPE (0=NONE / 1=ROM / 2=RAM / 3=RAM W-INIT)
1636 ; 49          00      WRITE PROTECT BITMAP HS SETTINGS
1637 ; 50          00      ACTION AFTER LOAD MODE PARAMETER (0=LOAD AUTO / 1=NEW / 2=RESET
1638 ;              3=JUMP TO LOAD_INIT ADDR / 4 MEMORY FILL WITH 00 / 5 NO ACTION
1639 ; 51          00      CARTRIDGE_HEADER_TYPE ( 01=LROS / 02=AROS / 03=TCC)
1640 ; 52          00      CARTRIDGE_INFO_LEN LOW (LROS=5 / AROS = 08 /TCC = 32)
1641 ; 53          00      CARTRIDGE_INFO_LEN HIGH
1642 ;-----
1643 ; TIMEX CARTRIDGE LROS / AROS HEADER (5/8 BYTES)
1644 ;-----
1645 ;
1646 ;
1647 ; 54          00      CARTRIDGE_00      NOT USED      LANG.TYPE
1648 ; 55          00      CARTRIDGE_01      TYPE          TYPE
1649 ; 56          00      CARTRIDGE_02      STARTING ADDR  STARTING ADDR
1650 ; 57          00      CARTRIDGE_03      STARTING ADDR  STARTING ADDR
1651 ; 58          00      CARTRIDGE_04      CHUNK SPECS    CHUNK SPECS
1652 ; 59          00      CARTRIDGE_05
1653 ; 60          00      CARTRIDGE_06      AUTOSTART
1654 ; 61          00      CARTRIDGE_07      RESERVED BYTES
1655 ;-----
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1681
1682 ;-----
1683 ;
1684 ; *****
1685 ; *** FIELDS ONLY AVAILABLES IN .TCC CARTRIDGE FORMAT ***
1686 ; *****
1687 ;
1688 ; 62          00      HEADER RELEASE VERSION
1689 ; 63          00      DESTINATION BANK NUMBER AFTER LOAD (0 = DEFAULT)
1690 ; 64          00      HS SETTINGS AFTER LOAD (0 = DISABLE 1 = ENABLE BANK)
1691 ; 65          00      AFTER LOAD INIT_ADDR LOW (USE ONLY ON ACTION = 3)
1692 ; 66          00      AFTER LOAD INIT_ADDR HIGH
1693 ; 67          00      VALID INTERRUPT ENABLE ROUTINE AFTER LOAD (0 = DISABLE / 1=ENABLE)
1694 ; 68          00      ASCII NAME
1695 ; 69          00      ASCII NAME
1696 ; 70          00      ASCII NAME
1697 ; 71          00      ASCII NAME
1698 ; 72          00      ASCII NAME
1699 ; 73          00      ASCII NAME
1700 ; 74          00      ASCII NAME
1701 ; 75          00      ASCII NAME
1702 ; 76          00      ASCII NAME
1703 ; 77          00      ASCII NAME
1704 ; 78          00      ASCII NAME
1705 ; 79          00      ASCII NAME
1706 ; 80          00      END OF STRING
1707 ; 81          00      SERIAL NUM_0
1708 ; 82          00      SERIAL NUM_1
1709 ; 83          00      SERIAL NUM_2
1710 ; 84          00      SERIAL NUM_3
1711 ; 85          00      CRC OF CARTRIDGE HEADER
1712 ;
1713 ; *****
1714 ; *** END FIELDS ONLY AVAILABLES IN .TCC CARTRIDGE FORMAT ***
1715 ; *****
1716 ;
1717 ;-----
1718 ; 86          6B      CRC FROM STORAGE
1719 ;-----
1720 ; 87          6B      CRC FROM TIMEX
1721 ;-----
1722 ; 88          40      WAIT/CONTINUE
1723 ;-----
1724 ; 89          29      REQUEST DATA BLOCK LOW TYPE ID (29 = CARTRIDGE RAW IMAGEN LOW)
1725 ;                   (OR ERROR CODE IF VALUE <20H OR >80H)
1726 ;-----
1727
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1751 ;-----
1752 ;
1753 ; COMMAND LDDOCK CARTRIDGE READ DATA BLOCK
1754 ;-----
1755 ; ***** LOAD DATA CHUNK 0 -> 6 LOAD DATA FROM 0000 -> DFFF ONLY BLOCK LOW ID = 02 *****
1756 ;-----
1757 ; 00          44      "E"      RESPONSE TO COMMAND
1758 ; 01          0A      BLK_LEN (RAW CARTRIDGE + 0AH)
1759 ; 02          A0
1760 ;-----
1761 ; 03          45      COMMAND ID (45=LDDOCK)
1762 ; 04          00      BANK # (00=DOCK FF=HOME ...)
1763 ; 05          00      SESION ID LOW
1764 ; 06          00      SESION ID HIGH
1765 ;-----
1766 ; 07          09      DATA PARAMETER ID ( 09 = CARTRIDGE RAW IMAGEN LOW)
1767 ; 08          01      COMMAND BLOCK NUMBER (00 = SINGLE COMMAND BLOCK/LAST MESSAGE / N
1768 ; 09          00      LOADING ADDR LOW
1769 ; 10          00      LOADING ADDR HIGH
1770 ; 11          0B      BLK_LEN LOW
1771 ; 12          00      BLK_LEN HIGH
1772 ;-----
1773 ;---56K MAXIMUN - BLOCK SIZE FOR 1ST BLOCK---
1774 ;-----
1775 ; 13          00      CARTRIDGE IMAGEN DATA
1776 ; 14          00
1777 ; .           .
1778 ; .           00
1779 ; .           00
1780 ;-----
1781 ; 57357        6B      CRC
1782 ;-----
1783 ; 57358        01      STATUS CODE
1784 ;-----
1785 ; 57359 40      WAIT/CONTINUE
1786 ;-----
1787 ; 57360        2A      REQUEST DATA BLOCK LOW TYPE ID (2A = CARTRIDGE RAW IMAGEN HIGH)
1788 ;                  (OR ERROR CODE IF VALUE <20H OR >80H)
1789 ;-----
1790 ;-----
1791 ;-----
1792 ; COMMAND LDDOCK CARTRIDGE READ DATA BLOCK
1793 ;-----
1794 ; ***** LOAD DATA CHUNK 7 LOAD DATA FROM E000 -> FFFF ONLY BLOCK HIGH ID = 03 *****
1795 ;-----
1796 ; 00          44      "E"      RESPONSE TO COMMAND
1797 ; 01          1C      BLK_LEN
1798 ; 02          00
1799 ;-----
1800 ; 03          45      COMMAND ID (45=LDDOCK)
1801 ; 04          00      BANK # (00=DOCK FF=HOME ...)
1802 ; 05          00      SESION ID LOW
1803 ; 06          00      SESION ID HIGH
1804 ;-----
1805 ; 07          2A      DATA PARAMETER ID ( 2A = CARTRIDGE RAW IMAGEN LOW)
1806 ; 08          00      COMMAND BLOCK NUMBER (00 = SINGLE COMMAND BLOCK/LAST MESSAGE / N
1807 ; 09          00      LOADING ADDR LOW
1808 ; 10          00      LOADING ADDR HIGH
1809 ; 11          0B      BLK_LEN LOW
1810 ; 12          00      BLK_LEN HIGH
1811 ;-----
1812 ;---8K MAXIMUN - BLOCK SIZE FOR HIGH BLOCK---
1813 ;-----
1814 ; 13          00      CARTRIDGE IMAGEN DATA
1815 ; 14          00
1816 ; .           .
1817 ; 8202        00
1818 ; 8203        00
1819 ;-----
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1821 ;-----
1822 ;
1823 ; 8205                6B      CRC
1824 ;-----
1825 ; 8206                01      STATUS CODE
1826 ;-----
1827 ; 8207  40            WAIT/CONTINUE
1828 ;-----
1829 ; 8208                27      NO MORE DATA REQUEST (27 = END)
1830 ;                          (OR ERROR CODE IF VALUE <20H OR >80H)
1831 ;-----
1832 ;*****
1833 ;
1834 ;-----
1835 ;
1836 ; COMMAND STATUS FROM ASSEMBLER                                COMMAND ID: 040H
1837 ;-----
1838 ; 00                44      "A" ; ASSEMBLER TYPE COMMAND
1839 ; 01                40      COMMAND ID (40=STATUS)
1840 ; 02                00      BANK # (00=DOCK FF=HOME ...)
1841 ; 03                00      SESION ID LOW
1842 ; 04                00      SESION ID HIGH
1843 ;-----
1844 ; 05                20      COMMAND PARAMETER ID (20= BINARY / RAW PARAM)
1845 ; 06                00      COMMAND BLOCK NUMBER (00 = SINGLE COMMAND BLOCK MESSAGE)
1846 ; 07                23      BLK_LEN LOW
1847 ; 08                00      BLK_LEN HIGH
1848 ;-----
1849 ; 09                6B      CRC
1850 ;-----
1851 ; 10      40            WAIT CONTINUE
1852 ;-----
1853 ; 11                01      STATUS CODE
1854 ;-----
1855 ; 12                44      "D"
1856 ; 13                23      BLK_LEN
1857 ; 14                00
1858 ;-----
1859 ; 15                00      SESION ID LOW
1860 ; 16                00      SESION ID HIGH
1861 ; 17                01      HEADER VERSION 1
1862 ; 18                00      CPU % USAGE
1863 ; 19                00      STORAGE FREE SPACE MB LOW
1864 ; 20                00      STORAGE FREE SPACE MB MEDIUM
1865 ; 21                00      STORAGE FREE SPACE MB HIGH
1866 ; 22                00      STORAGE IN USE SPACE MB LOW
1867 ; 23                00      STORAGE IN USE SPACE MB MEDIUM
1868 ; 24                00      STORAGE IN USE SPACE MB HIGH
1869 ; 25                00      MESSAGE NUMBER LOW
1870 ; 26                00      MESSAGE NUMBER MEDIUM
1871 ; 27                03      MESSAGE NUMBER HIGH
1872 ; 28                01      MEMORY FREE KB LOW
1873 ; 29                00      MEMORY FREE KB HIGH
1874 ; 30                00      ERROR COUNT LOW
1875 ; 31                00      ERROR COUNT HIGH
1876 ; 32                00      RUNNING SECONDS TIME LOW
1877 ; 33                00      RUNNING SECONDS TIME MED
1878 ; 34                00      RUNNING SECONDS TIME HIGH
1879 ;-----
1880 ;
1881 ; TAP FILE INFO
1882 ;-----
1882 ; 35                01      MOUNT TAP FILE FLAG (0=NO MOUNT/ 1=MOUNT)
1883 ; 36                00      TAP SIZE LOW
1884 ; 37                02      TAP SIZE MID
1885 ; 38                00      TAP SIZE HIGH
1886 ; 39                0A      TAP FILENAME LEN
1887 ;-----
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1891
1892 ;-----
1893 ;      DATA COMMAND ARGUMENT
1894 ;-----
1895 ; 40          46          "F"          ; Cartridge Filename on the SD
1896 ; 41          49          "I"
1897 ; 42          4C          "L"
1898 ; 43          45          "E"
1899 ; 44          30          "0"
1900 ; 45          31          "1"
1901 ; 46          2E          "."
1902 ; 47          54          "T"
1903 ; 48          41          "A"
1904 ; 49          50          "P"
1905 ;-----
1906 ; 50          6B          CRC
1907 ;-----
1908 ; 51          01          SEND STATUS CODE OF RECEIVED COMMAND
1909 ;-----
1910 ; 52      40          WAIT FOR CONTINUE TO FINISH
1911 ;-----
1912 ; 53          01          NO ERROR INDICATION (01 = OK)
1913 ;                      (OR ERROR CODE IF VALUE <20H OR >80H)
1914 ;-----
1915 ;*****
1916 ;-----
1917 ;-----
1918 ; COMMAND GETMPR          GET CURRENT MEMORY PAGE REGISTER VALUE
1919 ;-----
1920 ;-----
1921 ; 00          44          "A"          ; ASSEMBLER TYPE COMMAND
1922 ; 01          41          COMMAND ID (41=GETMPR)
1923 ; 02          00          BANK # (00=DOCK FF=HOME ...)
1924 ; 03          00          SESION ID LOW
1925 ; 04          00          SESION ID HIGH
1926 ;-----
1927 ; 05          01          COMMAND PARAMETER ID ( 00= NONE)
1928 ; 06          01          COMMAND BLOCK NUMBER (00 = LAST COMMAND BLOCK MESSAGE)
1929 ; 07          0B          BLK_LEN LOW
1930 ; 08          00          BLK_LEN HIGH
1931 ;-----
1932 ; 09          6B          CRC
1933 ;-----
1934 ; 10      40          WAIT CONTINUE
1935 ;-----
1936 ; 11          01          STATUS CODE
1937 ;-----
1938 ; 12          44          "D"
1939 ; 13          0B          BLK_LEN
1940 ; 14          00
1941 ;-----
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1960 TS2068-PICO TPI CMD SPECS AGO 18 2023 V:2.4 MACRO-80 3.44 09-Dec-81 PAGE 1-29
1961
1962 ;-----
1963 ; DATA COMMAND ARGUMENT
1964 ;-----
1965 ; 15 01 CURRENT MEMORY PAGE REGISTER VALUE
1966 ; 16 01 CURRENT HOME ROM MEMORY SOURCE ( 00 = NA / 01 = ROM / 02 = RAM)
1967 ; 17 00 CURRENT HOME ROM MAPPING SEGMENT (MA16 TO MA23)
1968 ; 18 01 CURRENT EXROM MEMORY SOURCE ( 00 = NA / 01 = ROM / 02 = RAM)
1969 ; 19 02 CURRENT EXROM MEMORY MAPPING SEGMENT (MA16 TO MA23)
1970 ; 20 02 CURRENT DOCK MEMORY SOURCE ( 00 = NA / 01 = ROM / 02 = RAM)
1971 ; 21 00 CURRENT DOCK MAPPING SEGMENT (MA16 TO MA23)
1972 ; 22 02 CURRENT DOCK MEMORY START LOW ADDR (0000 FOR LROS / 8000 FOR ARC)
1973 ; 23 02 CURRENT DOCK MEMORY START HIGH ADDR
1974 ; 24 02 CURRENT DOCK SIZE LOW
1975 ; 25 02 CURRENT DOCK SIZE HIGH
1976 ;-----
1977 ; 26 6B CRC
1978 ;-----
1979 ; 27 01 SEND STATUS CODE OF RECEIVED COMMAND
1980 ;-----
1981 ; 28 40 WAIT FOR CONTINUE TO FINISH
1982 ;-----
1983 ; 30 01 NO ERROR INDICATION (01 = OK)
1984 ; (OR ERROR CODE IF VALUE <20H OR >80H)
1985 ;-----
1986
1987 ;-----
1988
1989 ; *****
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2031
2032 LOAD "DIR *.DCK" DATA A\$()2033
2034 ;-----
2035 ; LOAD "DIR *.DCK" DATA A\$()2036 ;-----
2037 ; 00 BLOCK TYPE (00 = Header)
2038 ; 01 TADDR (01=LOAD)
2039 ; FF BANK2
2040 ; 69 SESSION ID
2041 ; 7F
2042 ; 63 MEMORY ADDRESS
2043 ; 6E
2044 ; 11 BLK_LEN
2045 ; 002046 ;-----
2047 ; F4 CRC FROM TIMEX2048 ;-----
2049 ; 01 STATUS CODE2050 ;-----
2051 ; 40 WAIT CONTINUE2052 ;-----
2053 ; 00 REQUESTED BLOCK TYPE (00=HEADER)2054 ;-----
2055 ; 00 BLOCK TYPE FROM FILE (00 = Header)2056 ;-----
2057 ; 02 TYPE ALPHANUM ARRAY

2058 ; 44 "D"

2059 ; 4F "O"

2060 ; 43 "C"

2061 ; 4B "K"

2062 ; 2E "."

2063 ; 44 "D"

2064 ; 41 "A"

2065 ; 54 "T"

2066 ; 20 " "

2067 ; 20 " "

2068 ; 45 BLOCK LEN (Low Byte)

2069 ; 01 BLOCK LEN (High Byte)

2070 ; 22 NOT USED

2071 ; C1 CHAR TYPE + LETTER "A"

2072 ; 41 "A"

2073 ; 80 NOT USED

2074 ;-----
2075 ; 18 CRC VALUE FROM FILE2076 ;-----
2077 ; 00 CRC VALUE FROM TIMEX2078 ;-----
2079 ; 40 WAIT CONTINUE2080 ;-----
2081 ; 01 STATUS CODE2082 ;-----
2083 ; FF BLOCK TYPE (FF = Header)

2084 ; 01 TADDR (01=LOAD)

2085 ; FF BANK # (00=DOCK FF=HOME ...)

2086 ; 69 SESSION ID LOW

2087 ; 7F SESSION ID HIGH

2088 ; 00 MEMORY ADDRESS LOW

2089 ; 00 MEMORY ADDRESS HIGH

2090 ; 45 BLK_LEN LOW

2091 ; 01 BLK_LEN HIGH

2092 ;-----
2093 ; F4 CRC FROM TIMEX2094 ;-----
2095 ; 01 STATUS CODE2096 ;-----
2097 ; 40 WAIT CONTINUE2098 ;-----
2099
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2101
2102 ;-----
2103 ;                FF                REQUESTED BLOCK TYPE (FF=BLOCK)
2104 ;-----
2105 ;                FF                BLOCK TYPE FROM FILE (FF = BLOCK)
2106 ;-----
2107 ;                02                CHARACTER ARRAY
2108 ;                14                MAX FILES LOW
2109 ;                00                MAX FILES HIGH
2110 ;                10                RECORD LEN LOW
2111 ;                00                RECORD LEN HIGH
2112 ;-----
2113 ;                                31 38 20      18
2114 ; 0020 20 20 20 20 20 20 20 20 20 20 20 20 20 32 32 32      16
2115 ; 0030 32 2E 64 63 6B 20 20 20 20 20 20 20 20 33 33 33      3333.dck
2116 ; 0040 33 2E 64 63 6B 20 20 20 20 20 20 20 20 34 34 34      4444.dck
2117 ; 0050 34 2E 64 63 6B 20 20 20 20 20 20 20 20 35 35 35      5555.dck
2118 ; 0060 35 2E 64 63 6B 20 20 20 20 20 20 20 20 36 36 36      6666.dck
2119 ; 0070 36 2E 64 63 6B 20 20 20 20 20 20 20 20 37 37 37      7777.dck
2120 ; 0080 37 2E 64 63 6B 20 20 20 20 20 20 20 20 38 38 38      8888.dck
2121 ; 0090 38 2E 64 63 6B 20 20 20 20 20 20 20 20 39 39 39      9999.dck
2122 ; 00A0 39 2E 64 63 6B 20 20 20 20 20 20 20 20 31 30 31      10101010.dck
2123 ; 00B0 30 31 30 31 30 2E 64 63 6B 20 20 20 20 31 31 31      11111111.dck
2124 ; 00C0 31 31 31 31 31 2E 64 63 6B 20 20 20 20 31 32 31      12121212.dck
2125 ; 00D0 32 31 32 31 32 2E 64 63 6B 20 20 20 20 31 33 31      13131313.dck
2126 ; 00E0 33 31 33 31 33 2E 64 63 6B 20 20 20 20 31 34 31      14141414.dck
2127 ; 00F0 34 31 34 31 34 2E 64 63 6B 20 20 20 20 31 35 31      15151515.dck
2128 ; 0100 35 31 35 31 35 2E 64 63 6B 20 20 20 20 31 36 31      16161616.dck
2129 ; 0110 36 31 36 31 36 2E 64 63 6B 20 20 20 20 31 37 31      17171717.dck
2130 ; 0120 37 31 37 31 37 2E 64 63 6B 20 20 20 20 31 38 31      18181818.dck
2131 ; 0130 38 31 38 31 38 2E 64 63 6B 20 20 20 20 31 39 31      19191919.dck
2132 ; 0140 39 31 39 31 39 2E 64 63 6B 20 20 20 20 32 30 32      20202020.dck
2133 ; 0150 30 32 30 32 30 2E 64 63 6B 20 20 20 20
2134 ;-----
2135 ;                F7                CRC FROM FILE
2136 ;-----
2137
2138 ;-----
2139 ;                F4                CRC FROM TIMEX
2140 ;-----
2141 ;                01                STATUS CODE
2142 ;-----
2143 ;                40                WAIT CONTINUE
2144 ;-----
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2173 *****
2174 **      THE FOLLOWING FUNCTIONS REQUIRE A 64K DOCK BANK      **
2175 *****
2176 ** SAVE "TPI:SNAPSHOT" **
2177 ** **
2178 ** CP/M CBIOS MESSAGES **
2179 ** **
2180 *****
2181 **      * P R E L I M I N A R I N F O R M A T I O N *      **
2182 *****
2183

```

2183

2184 SEND THE FILENAME TO THE STORAGE DEVICE WITH THE FOLLOWING ASSEMBLER COMMAND

2185

2186 SET FILENAME TO USE IN NEXT SAVE/LOAD "Timex Memory Snapshot" (.TMS)

2187

The Snapshot routines run at 3400H in the DOCK RAM Bank

2188

2189 USE THE SAVE COMMAND TO SET-UP THE FILENAME BEFORE

2190

USE THE SNAPSHOT SAVE/LOAD BLOCK COMMANDS

2191

2192 SAVE "TPI:SNAPSHOT"

2193

2194

2195 -----

2196

CO RD-15 WR-14 RD-14

2197

```

=====
2197 00      0x42      "A" (BASIC COMND)
2198 01      0x47      COMMAND ID (47=SNSNAP)
2199 02      0xFF      BANK # (00=DOCK FF=HOME ...)
2200 03      0x00      SESION ID LOW
2201 04      0x00      SESION ID HIGH
2202 ;-----

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2240 TS2068-PICO TPI CMD SPECS   AGO 18 2023 V:2.4   MACRO-80 3.44   09-Dec-81 PAGE 1-33
2241
2242 FUNCTION LOAD SNAPSHOT BLOCK IMAGE FROM SD           FUNCTION SAVE SNAPSHOT BLOCK IMAGE ON SD
2243
2244 -----
2245 CO RD-15      WR-14      RD-14                      CO RD-15      WR-14      RD-14
2246 =====
2247 00            0x42      "A" (ASSEMBLER COMND)      00            0x42      "A" (ASSEMBLER COMND)
2248 01            0x48      LOAD SNAP (TADDR)          01            0x49      SAVE SNAP (TADDR)
2249 02            0xFF      BANK NUMBER                02            0xFF      BANK NUMBER
2250 03            0x01      SESION ID LOW               03            0x01      SESION ID LOW
2251 04            0x01      SESION ID HIGH              04            0x00      SESION ID HIGH
2252 -----
2253 05            0x2C      COM PMR ID (2C= H.REG)      05            0x2C      COM PMR ID (2C= H.REG)
2254 06            0x03      CMND BLK NUMB              06            0x03      CMND BLK NUMB
2255 07            0x0B      BLK_LEN LOW                07            0x0B      BLK_LEN LOW
2256 08            0x00      BLK_LEN HIGH                08            0x00      BLK_LEN HIGH
2257 -----
2258 09            0x2E      CRC FROM TIMEX              09            0x27      CRC FROM TIMEX
2259 -----
2260 10            0x01      Status Code                 10            0x01      Status Code
2261 -----WAIT EXECUTION-----                      -----WAIT EXECUTION-----
2262 11 0x40      CONTINUE FLAG                          11 0x40      CONTINUE FLAG
2263 -----
2264 12            0x44      "D" (DATA BLOCK CMD)        12            0x44      "D" (DATA BLOCK CMD)
2265 13            0x00      SNAP_TOTAL_BLOCK_LEN L      13            0x00      SNAP_TOTAL_BLOCK_LEN L
2266 14            0xC0      SNAP_TOTAL_BLOCK_LEN H      14            0x00      SNAP_TOTAL_BLOCK_LEN H
2267 -----
2268 15            0x00      HEADER_BLOCK_LEN L          15            0x00      HEADER_BLOCK_LEN L
2269 16            0x00      HEADER_BLOCK_LEN H          16            0x00      HEADER_BLOCK_LEN H
2270 17            0x01      HEADER_VERSION_ID           17            0x01      HEADER_VERSION_ID
2271 -----
2272 18            0x00      SNAP_OFST_Z80REG            18            0x00      SNAP_OFST_Z80REG
2273 19            0x00      SNAP_HEADER_LEN             19            0x00      SNAP_HEADER_LEN
2274 -----
2275 20            0x00      MAIN_ROM_ID                 20            0x00      MAIN_ROM_ID
2276 21            0x00      ROM_M393_IF1_FLAG           21            0x00      ROM_M393_FLAG
2277 22            0x00      ROM_M393_IF1_MAP            22            0x00      ROM_M393_IF1_MAP
2278 23            0x00      ROM_SIG_0                  23            0x00      ROM_SIG_0
2279 24            0x00      ROM_SIG_1                  24            0x00      ROM_SIG_1
2280 25            0x00      ROM_SIG_2                  25            0x00      ROM_SIG_2
2281 26            0x00      ROM_SIG_3                  26            0x00      ROM_SIG_3
2282 -----
2283 27            0x00      MACH_PORT_FF                27            0x00      MACH_PORT_FF
2284 28            0x00      MACH_PORT_FE                28            0x00      MACH_PORT_FE
2285 29            0x00      MACH_PORT_F4                29            0x00      MACH_PORT_F4
2286 30            0x00      MACH_PORT_EF                30            0x00      MACH_PORT_EF
2287 -----
2288 31            0x00      MACH_AY3_FLAG                31            0x00      MACH_AY3_FLAG
2289 32            0x00      MACH_AY3_DATA_PORT L        32            0x00      MACH_AY3_DATA_PORT L
2290 33            0x00      MACH_AY3_DATA_PORT H        33            0x00      MACH_AY3_DATA_PORT H
2291 34            0x00      MACH_AY3_ADDR_PORT L        34            0x00      MACH_AY3_ADDR_PORT L
2292 35            0x00      MACH_AY3_ADDR_PORT H        35            0x00      MACH_AY3_ADDR_PORT H
2293 36            0x00      MACH_AY3_DA_00              36            0x00      MACH_AY3_DA_00
2294 37            0x00      MACH_AY3_DA_01              37            0x00      MACH_AY3_DA_01
2295 38            0x00      MACH_AY3_DA_02              38            0x00      MACH_AY3_DA_02
2296 39            0x00      MACH_AY3_DA_03              39            0x00      MACH_AY3_DA_03
2297 40            0x00      MACH_AY3_DA_04              40            0x00      MACH_AY3_DA_04
2298 41            0x00      MACH_AY3_DA_05              41            0x00      MACH_AY3_DA_05
2299 42            0x00      MACH_AY3_DA_06              42            0x00      MACH_AY3_DA_06
2300 43            0x00      MACH_AY3_DA_07              43            0x00      MACH_AY3_DA_07
2301 44            0x00      MACH_AY3_DA_08              44            0x00      MACH_AY3_DA_08
2302 45            0x00      MACH_AY3_DA_09              45            0x00      MACH_AY3_DA_09
2303 46            0x00      MACH_AY3_DA_10              46            0x00      MACH_AY3_DA_10
2304 47            0x00      MACH_AY3_DA_11              47            0x00      MACH_AY3_DA_11
2305 -----
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2310	-----						-----					
2311	48		0x00	MACH_AY3_DA_12	48		0x00	MACH_AY3_DA_12				
2312	49		0x00	MACH_AY3_DA_13	49		0x00	MACH_AY3_DA_13				
2313	50		0x00	MACH_AY3_DA_14	50		0x00	MACH_AY3_DA_14				
2314	51		0x00	MACH_AY3_DA_15	51		0x00	MACH_AY3_DA_15				
2315	52		0x00	MACH_DOCK_ROM_FLAG	52		0x00	MACH_DOCK_ROM_FLAG				
2316	53		0x00	MACH_DOCK_HS	53		0x00	MACH_DOCK_HS				
2317	54		0x00	MACH_DOCK_AROS_FLAG	54		0x00	MACH_DOCK_AROS_FLAG				
2318	55		0x00	MACH_DOCK_LROS_FLAG	55		0x00	MACH_DOCK_LROS_FLAG				
2319	56		0x00	MACH_DOCK_RAM_FLAG	56		0x00	MACH_DOCK_RAM_FLAG				
2320	57		0x00	MACH_DOCK_SIZE_KB	57		0x00	MACH_DOCK_SIZE_KB				
2321	58		0x00	MACH_EXROM_FLAG	58		0x00	MACH_EXROM_FLAG				
2322	59		0x00	MACH_EXPB_QTY	59		0x00	MACH_EXPB_QTY				
2323	60		0x01	MEM_TOTAL_BLOCKS_QTY	60		0x00	MEM_TOTAL_BLOCKS_QTY				
2324	-----						-----					
2325	61		0x01	CPU_ID	61		0x00	CPU_ID				
2326	62		0x01	CPU_MODEL	62		0x00	CPU_MODEL				
2327	63		0xC8	CPU_KHZ L	63		0x00	CPU_KHZ L				
2328	64		0x0D	CPU_KHZ H	64		0x00	CPU_KHZ H				
2329	-----						-----					
2330	65		0x00	CPU_REG_FLAGS	65		0x00	CPU_REG_FLAGS				
2331	66		0x00	CPU_REG_A	66		0x00	CPU_REG_A				
2332	67		0x00	CPU_REG_C	67		0x00	CPU_REG_C				
2333	68		0x00	CPU_REG_B	68		0x00	CPU_REG_B				
2334	69		0x00	CPU_REG_E	69		0x00	CPU_REG_E				
2335	70		0x00	CPU_REG_D	70		0x00	CPU_REG_D				
2336	71		0x00	CPU_REG_L	71		0x00	CPU_REG_L				
2337	72		0x00	CPU_REG_H	72		0x00	CPU_REG_H				
2338	73		0x00	CPU_FLAGS_ALT	73		0x00	CPU_FLAGS_ALT				
2339	74		0x00	CPU_REG_A_ALT	74		0x00	CPU_REG_A_ALT				
2340	75		0x00	CPU_REG_C_ALT	75		0x00	CPU_REG_C_ALT				
2341	76		0x00	CPU_REG_B_ALT	76		0x00	CPU_REG_B_ALT				
2342	77		0x00	CPU_REG_E_ALT	77		0x00	CPU_REG_E_ALT				
2343	78		0x00	CPU_REG_D_ALT	78		0x00	CPU_REG_D_ALT				
2344	79		0x00	CPU_REG_H_ALT	79		0x00	CPU_REG_H_ALT				
2345	80		0x00	CPU_REG_L_ALT	80		0x00	CPU_REG_L_ALT				
2346	81		0x00	CPU_REG_IX L	81		0x00	CPU_REG_IX L				
2347	82		0x00	CPU_REG_IX H	82		0x00	CPU_REG_IX H				
2348	83		0x00	CPU_REG_IY L	83		0x00	CPU_REG_IY L				
2349	84		0x00	CPU_REG_IY H	84		0x00	CPU_REG_IY H				
2350	85		0x00	CPU_REG_SP L	85		0x00	CPU_REG_SP L				
2351	86		0x00	CPU_REG_SP H	86		0x00	CPU_REG_SP H				
2352	87		0x00	CPU_REG_PC L	87		0x00	CPU_REG_PC L				
2353	88		0x00	CPU_REG_PC H	88		0x00	CPU_REG_PC H				
2354	89		0x00	CPU_REG_I	89		0x00	CPU_REG_I				
2355	90		0x00	CPU_REG_R	90		0x00	CPU_REG_R				
2356	91		0x00	CPU_REG_R7	91		0x00	CPU_REG_R7				
2357	92		0x00	CPU_REG_MEMPTR	92		0x00	CPU_REG_MEMPTR				
2358	93		0x00	CPU_REG_IFF1	93		0x00	CPU_REG_IFF1				
2359	94		0x00	CPU_REG_IFF2	94		0x00	CPU_REG_IFF2				
2360	95		0x00	CPU_IM_MODE	95		0x00	CPU_IM_MODE				
2361	-----						-----					
2362	96		0x00	SNAP_MODE	96		0x00	SNAP_MODE				
2363	97		0x00	SNAP_LOAD_ADDR L	97		0x00	SNAP_SAVE_ADDR L				
2364	98		0x00	SNAP_LOAD_ADDR H	98		0x00	SNAP_SAVE_ADDR H				
2365	99		0x00	CRC_FILE_HEADER	99		0x00	CRC_FILE_HEADER				
2366	-----						-----					
2367	100		0x77	CRC FROM TIMEX	100		0x7A	CRC FROM TIMEX				
2368	-----						-----					
2369	101		0x01	Status Code	101		0x01	Status Code				
2370	-----WAIT EXECUTION-----						-----WAIT EXECUTION-----					
2371	102	0x40		CONTINUE FLAG	102	0x40		CONTINUE FLAG				
2372	-----						-----					
2373	103		0x2D	REQ PARAM ID=2D H.DATA	103		0x2D	SET PARAM ID=2D H.DATA				
2374	(OR ERROR CODE IF VALUE <20H OR >80H)						(OR ERROR CODE IF VALUE <20H OR >80H)					
2375	-----						-----					
2376												
2377												
2378												
2379												
2380												
2380												
2380												
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2380 TS2068-PICO TPI CMD SPECS   AGO 18 2023 V:2.4   MACRO-80 3.44   09-Dec-81 PAGE 1-35
2381 -----
2382 CO RD-15   WR-14   RD-14
2383 =====
2384 00          0x42      "A" (ASSEMBLER COMND)  00          0x42      "A" (ASSEMBLER COMND)
2385 01          0x48      LOAD SNAP (TADDR)      01          0x49      SAVE SNAP (TADDR)
2386 02          0xFF      BANK NUMBER            02          0xFF      BANK NUMBER
2387 03          0x01      SESION ID LOW           03          0x01      SESION ID LOW
2388 04          0x01      SESION ID HIGH          04          0x00      SESION ID HIGH
2389 -----
2390 05          0x2D      COM PMR ID (2D= H.DATA) 05          0x2D      COM PMR ID (2D= H.DATA)
2391 06          0x03      CMND BLK NUMB          06          0x03      CMND BLK NUMB
2392 07          0x0B      BLK_LEN LOW            07          0x0B      BLK_LEN LOW
2393 08          0x00      BLK_LEN HIGH           08          0x00      BLK_LEN HIGH
2394 -----
2395 09          0x2E      CRC FROM TIMEX          09          0x27      CRC FROM TIMEX
2396 -----
2397 10          0x01      Status Code             10          0x01      Status Code
2398 -----WAIT EXECUTION-----
2399 11 0x40      CONTINUE FLAG                   11 0x40      CONTINUE FLAG
2400 -----
2401 105         0x44      "D" (DATA BLOCK CMD)    103         0x44      "D" (DATA BLOCK CMD)
2402 -----
2403 106         0x00      MEM_BLOCK1_BANK         104         0x00      MEM_BLOCK1_BANK
2404 107         0x00      MEM_HOME_START_ADDR L   105         0x00      MEM_HOME_START_ADDR L
2405 108         0x00      MEM_HOME_START_ADDR H   106         0x00      MEM_HOME_START_ADDR H
2406 109         0x00      MEM_HOME_LEN L          107         0x00      MEM_HOME_LEN L
2407 110         0x00      MEM_HOME_LEN H          108         0x00      MEM_HOME_LEN H
2408 -----
2409 111         0x00      MEM_HOME_DATA1_4000     111         0x00      MEM_HOME_DATA1_4000
2410 112         0x00      MEM_HOME_DATA1_4001     112         0x00      MEM_HOME_DATA1_4001
2411 .
2412 49262       0x00      MEM_HOME_DATA1_FFFF     49262       0x00      MEM_HOME_DATA1_FFFF
2413 -----
2414 49261       0x00      CRC FROM TIMEX          49261       0x00      CRC FROM TIMEX
2415 -----
2416 49262       0x01      Send Status Code OK     49262       0x01      Get Status Code
2417 -----WAIT EXECUTION-----
2418 49263 0x40   CONTINUE FLAG                   49263 0x40   CONTINUE FLAG
2419 -----
2420 49264       0x2E      REQ PARAM ID=2C H.REG.  49264       0x2E      SET PARAM ID=2C H.REG.
2421         (OR ERROR CODE IF VALUE <20H OR >80H) (OR ERROR CODE IF VALUE <20H OR >80H)
2422 -----
2423 -----
2424 -----
2425 CO RD-15   WR-14   RD-14
2426 =====
2427 00          0x42      "A" (ASSEMBLER COMND)  00          0x42      "A" (ASSEMBLER COMND)
2428 01          0x48      LOAD SNAP (TADDR)      01          0x49      SAVE SNAP (TADDR)
2429 02          0xFF      BANK NUMBER            02          0xFF      BANK NUMBER
2430 03          0x01      SESION ID LOW           03          0x01      SESION ID LOW
2431 04          0x01      SESION ID HIGH          04          0x00      SESION ID HIGH
2432 -----
2433 05          0x2E      COM PMR ID (2C= H.REG)  05          0x2E      COM PMR ID (2C= H.REG)
2434 06          0x03      CMND BLK NUMB          06          0x03      CMND BLK NUMB
2435 07          0x0B      BLK_LEN LOW            07          0x0B      BLK_LEN LOW
2436 08          0x00      BLK_LEN HIGH           08          0x00      BLK_LEN HIGH
2437 -----
2438 09          0x2E      CRC FROM TIMEX          09          0x27      CRC FROM TIMEX
2439 -----
2440 10          0x01      Status Code             10          0x01      Status Code
2441 -----WAIT EXECUTION-----
2442 11 0x40      CONTINUE FLAG                   11 0x40      CONTINUE FLAG
2443 -----
2444 -----
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2450	TS2068-PICO TPI CMD SPECS	AGO 18 2023 V:2.4	MACRO-80 3.44	09-Dec-81	PAGE 1-36
2451					
2452					
2453	49265	0x44	"D" (DATA BLOCK CMD)	49264	0x44 "D" (DATA BLOCK CMD)
2454					
2455	49267	0x00	MEM_BLOCK2_BANK	49265	0x00 MEM_BLOCK2_BANK
2456	49268	0x00	MEM_DOCK_START_ADDR L	49266	0x00 MEM_DOCK_START_ADDR L
2457	49269	0x00	MEM_DOCK_START_ADDR H	49267	0x00 MEM_DOCK_START_ADDR H
2458	49270	0x00	MEM_DOCK_LEN L	49268	0x00 MEM_DOCK_LEN L
2459	49271	0x00	MEM_DOCK_LEN H	49269	0x00 MEM_DOCK_LEN H
2460					
2461	49272	0x00	MEM_DOCK_DATA2_4000	49270	0x00 MEM_DOCK_DATA2_4000
2462	49273	0x00	MEM_DOCK_DATA2_4001	49271	0x00 MEM_DOCK_DATA2_4001
2463	.			.	
2464	98423	0x00	MEM_DOCK_DATA3_FFFF	98421	0x00 MEM_DOCK_DATA3_FFFF
2465					
2466	98422	0x77	CRC FROM PICO	98422	0x7A CRC FROM TIMEX
2467					
2468	98423	0x01	Status Code	98423	0x01 Status Code
2469					
2470	98424 0x40		CONTINUE FLAG	98424 0x40	CONTINUE FLAG
2471					
2472	49264	0x2F	REQ PARM (2F EXP.BANK)	49264	0x2F REQ PARM (2F EXP.BANK)
2473			(OR ERROR CODE IF VALUE <20H OR >80H)		(OR ERROR CODE IF VALUE <20H OR >80H)
2474					
2475					
2476					
2477	CO RD-15	WR-14	RD-14	CO RD-15	WR-14 RD-14
2478					
2479	00	0x42	"A" (ASSEMBLER COMND)	00	0x42 "A" (ASSEMBLER COMND)
2480	01	0x48	LOAD SNAP (TADDR)	01	0x49 SAVE SNAP (TADDR)
2481	02	0xFF	BANK NUMBER	02	0xFF BANK NUMBER
2482	03	0x01	SESSION ID LOW	03	0x01 SESSION ID LOW
2483	04	0x01	SESSION ID HIGH	04	0x00 SESSION ID HIGH
2484					
2485	05	0x2F	COM PMR ID(2F EXP.BANK)	05	0x2F COM PMR ID (2F= H.REG)
2486	06	0x03	CMND BLK NUMB	06	0x03 CMND BLK NUMB
2487	07	0x0B	BLK_LEN LOW	07	0x0B BLK_LEN LOW
2488	08	0x00	BLK_LEN HIGH	08	0x00 BLK_LEN HIGH
2489					
2490	09	0x2E	CRC FROM TIMEX	09	0x27 CRC FROM TIMEX
2491					
2492	10	0x01	Status Code	10	0x01 Status Code
2493					
2494	11 0x40		CONTINUE FLAG	11 0x40	CONTINUE FLAG
2495					
2496					
2497					
2498	98425	0x44	"D" (DATA BLOCK CMD)	98425	0x44 "D" (DATA BLOCK CMD)
2499					
2500	98426	0x00	MEM_BLOCK3_BANK	98426	0x00 MEM_BLOCK3_BANK
2501	98427	0x00	MEM_EXPB_START_ADDR L	98427	0x00 MEM_EXPB_START_ADDR L
2502	98428	0x00	MEM_EXPB_START_ADDR H	98428	0x00 MEM_EXPB_START_ADDR H
2503	98429	0x00	MEM_EXPB_LEN L	98429	0x00 MEM_EXPB_LEN L
2504	98430	0x00	MEM_EXPB_LEN H	98430	0x00 MEM_EXPB_LEN H
2505					
2506	98431	0x00	MEM_EXPB_DATA3_4000	98431	0x00 MEM_EXPB_DATA3_4000
2507	98432	0x00	MEM_EXPB_DATA3_4001	98432	0x00 MEM_EXPB_DATA3_4001
2508	.			.	
2509	147582	0x00	MEM_EXPB_DATA3_FFFF	147582	0x00 MEM_EXPB_DATA3_FFFF
2510					
2511	147583	0x44	CRC FROM PICO	147583	0x44 CRC FROM TIMEX
2512					
2513	147584	0x01	Status Code	147584	0x01 Status Code
2514					
2515	147585 0x40		CONTINUE FLAG	147585 0x40	CONTINUE FLAG
2516					
2517	147586	0x01	NO ERROR OK	49264	0x01 NO ERROR OK
2518			(OR ERROR CODE IF VALUE <20H OR >80H)		(OR ERROR CODE IF VALUE <20H OR >80H)
2519					
2520					
2520					
2520					
2520					
2520					
2520					

2521

2522

CP/M 2.2 CBIOS MESSAGES

2523

CBIOS BOOT PRECESS MESSAGE

2524

2525

2526

CO RD-15 WR-14 RD-14

2527

00 0x43 "C" (CP/M COMMAND)

2528

01 0x00 CBIOS FUNCT

2529

02 0x00 CPU REG A

2530

03 0x00 CPU REG C

2531

04 0x00 CPU REG B

2532

05 0x00 CPU REG E

2533

06 0x00 CPU REG D

2534

07 0x00 CPU REG L

2535

08 0x00 CPU REG H

2536

2537

09 0x43 CRC FROM TIMEX

2538

2539

11 0x01 Status Code

2540

-----WAIT EXECUTION-----

2541

10 0x40 CONTINUE FLAG

2542

2543

2544

2545

CBIOS READ SECTOR MESSAGE

2546

2547

CO RD-15 WR-14 RD-14

2548

00 0x43 "C" (CP/M COMMAND)

2549

01 0x0D CBIOS FUNCT

2550

02 0x00 CPU REG A

2551

03 0x00 CPU REG C

2552

04 0x00 CPU REG B

2553

05 0x00 CPU REG E

2554

06 0x00 CPU REG D

2555

07 0x00 CPU REG L

2556

08 0x00 CPU REG H

2557

2558

09 0x4E CRC FROM TIMEX

2559

2560

10 0x01 Status Code

2561

-----WAIT EXECUTION-----

2562

11 0x40 CONTINUE FLAG

2563

2564

12 0x44 "D" (DATA BLOCK CMD)

2565

13 0x00 COMND LEN L

2566

14 0x1B COMND LEN H

2567

15 0x00 CBIOS FUNCT RESPONSE

2568

16 0x00 CPU REG A

2569

17 0x00 CPU REG C

2570

18 0x00 CPU REG B

2571

19 0x00 CPU REG E

2572

20 0x00 CPU REG D

2573

21 0x0F CPU REG L

2574

22 0x0F CPU REG H

2575

2576

23 0x0F SECTOR BYTE 00

2577

24 0x0F SECTOR BYTE 01

2578

2579

149 0x0F SECTOR BYTE 7E

2580

150 0x0F SECTOR BYTE 7F

2581

2582

151 0x93 CRC FROM PICO

2583

-----WAIT EXECUTION-----

2584

152 0x01 Status Code

2585

2586

153 0x40 CONTINUE FLAG

2587

2588

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CBIOS PRINT CONSOLE "A" CHARACTER

CO RD-15 WR-14 RD-14

00 0x43 "C" (CP/M COMMAND)

01 0x04 CBIOS FUN

02 0x00 CPU REG A

03 0x41 CPU REG C

04 0x00 CPU REG B

05 0x00 CPU REG E

06 0x00 CPU REG D

07 0x00 CPU REG L

08 0x00 CPU REG H

09 0x06 CRC FROM TIMEX

11 0x01 Status Code

-----WAIT EXECUTION-----

10 0x40 CONTINUE FLAG

CBIOS WRITE SECTOR MESSAGE

CO RD-15 WR-14 RD-14

00 0x43 "C" (CP/M COMMAND)

01 0x0E CBIOS FUN

02 0x00 CPU REG A

03 0x00 CPU REG C (DEBLOCK CODE)

04 0x00 CPU REG B

05 0x00 CPU REG E

06 0x00 CPU REG D

07 0x00 CPU REG L

08 0x00 CPU REG H

09 0x0C CRC FROM TIMEX

10 0x01 Status Code

-----WAIT EXECUTION-----

11 0x40 CONTINUE FLAG

12 0x44 "D" (DATA BLOCK CMD)

13 0x0A COMND LEN L

14 0x00 COMND LEN H

15 0x54 CBIOS FUNCT RESPONSE

16 0x50 CPU REG A

17 0x49 CPU REG C

18 0x3A CPU REG B

19 0x44 CPU REG E

20 0x45 CPU REG D

21 0x4C CPU REG L

22 0x45 CPU REG H

23 0x54 SECTOR BYTE 00

24 0x45 SECTOR BYTE 01

23 0x54 SECTOR BYTE 7E

24 0x45 SECTOR BYTE 7F

151 0x20 CRC FROM TIMEX

-----WAIT EXECUTION-----

152 0x01 Status Code

153 0x40 CONTINUE FLAG

2591 *****

2592

2593

2594 MANDATORY CP/M CBIOS FUNCTION MAP

2595 -----

2596	V:2.2	V:3.0	FUNC #		; Description
2597					
2598	YES	YES	00	BOOT	; Perform cold start initialization
2599	YES	YES	01	WBOOT	; Perform warm start initialization
2600	YES	YES	02	CONST	; Check for console input character ready
2601	YES	YES	03	CONIN	; Read Console Character in
2602	YES	YES	04	CONOUT	; Write Console Character out
2603	YES	YES	05	LIST	; Write List Character out
2604	YES	YES	06	PUNCH	; Write Auxiliary Output Character (USED AS ALT STORAGE DEVICE)
2605	YES	YES	07	READER	; Read Auxiliary Input Character (USED AS ALT STORAGE DEVICE)
2606	YES	YES	08	HOME	; Move to Track 00 on Selected Disk
2607	YES	YES	09	SELDISK	; Select Disk Drive
2608	YES	YES	0A	SETTRK	; Set Track Number
2609	YES	YES	0B	SETSEC	; Set Sector Number
2610	NO	NO	0C	SETDMA	; Set DMA Address
2611	YES	YES	0D	READ	; Read Specified Sector
2612	YES	YES	0E	WRITE	; Write Specified Sector
2613	YES	YES	0F	LISTST	; Return List Status
2614	NO	NO	10	SECTTRAN	; Translate Logical to Physical Sector
2615	-----CP/M 3-----				
2616	NO	YES	11	CONOST	; Return Output Status of Console
2617	NO	YES	12	AUXIST	; Return Input Status of Aux. Port
2618	NO	YES	13	AUXOST	; Return Output Status of Aux. Port
2619	NO	YES	14	DEVTBL	; Return Address of Char. I/O Table
2620	NO	YES	15	DEVINI	; Initialize Char. I/O Devices
2621	NO	YES	16	DRVTBL	; Return Address of DiskDrive Table
2622	NO	YES	17	MULTIO	; Set Number of Logically Consecutive sectors to be RD or WR
2623	NO	YES	18	FLUSH	; Force Physical Buffer Flushing for user-supported deblocking
2624	NO	YES	19	MOVE	; Memory to Memory Move
2625	NO	YES	1A	TIME	; Time Set/Get signal Clock
2626	NO	YES	1B	SELMEM	; Select memory bank
2627	NO	YES	1C	SETBNK	; Select bank for DMA operation
2628	NO	YES	1D	XMOVE	; Set Bank When a Buffer is in a Bank other than 0 or 1
2629	NO	YES	1E	USERF	; Reserved for System Implementor
2630	NO	YES	1F	RESERV1	; Reserved
2631	NO	YES	20	RESERV2	; Reserved
2632	NO	YES	21	RESERV3	; Reserved
2633	NO	YES	22	RESERV4	; Reserved
2634	NO	YES	23	RESERV5	; Reserved
2635	NO	YES	24	RESERV6	; Send STORAGE HWRD CMND (PHY,INIT CONFIG)
2636	NO	YES	25	RESERV6	; Set TERMINAL HWRD CMND
2637	NO	YES	26	RESERV6	; Set CPM CONFIG CMND

2638 *****

2639

2640 (C) Gustavo Pane

2641 Aug 2023

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2644 END OF DOC REV: 2.4

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